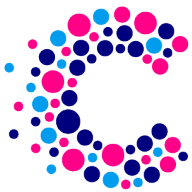
Multiple Comparisons

Luca Porcu & Chandra Chilamakuri (Bioinformatics Core)

Cancer Research UK Cambridge Institute, University of Cambridge

26 October 2026, 09:30–12:30

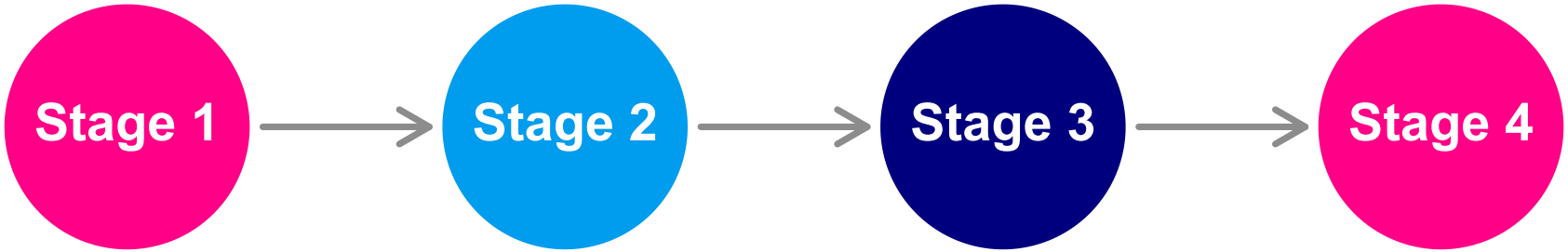


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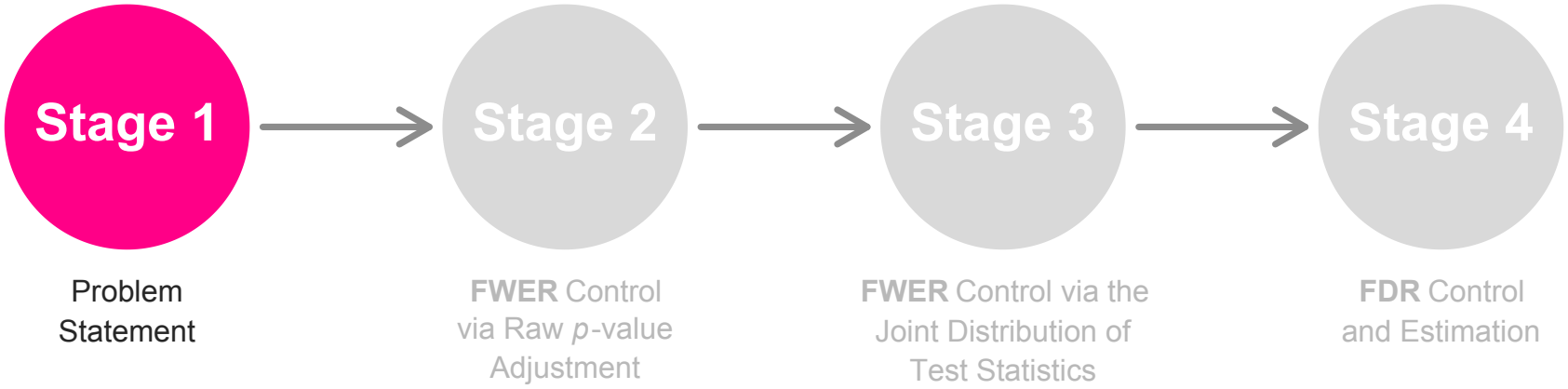
Roadmap



FWER: Family-Wise Error Rate

FDR: False Discovery Rate

Roadmap



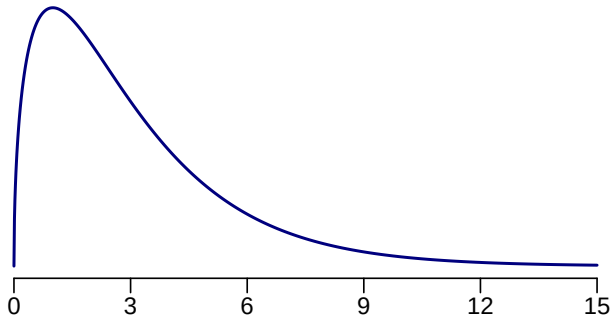
FWER: Family-Wise Error Rate

FDR: False Discovery Rate

Statistical concepts used for data-driven decisions

1 **Population** (i.e. statistical model) $\mathcal{P}_\theta$ → 2 A **hypothesis** (H) is a statement about $\mathcal{P}_\theta$

e.g. $\theta = 5$
$\theta \leq 10$
$\theta_1 = \theta_2$ (i.e. Pop. 1 = Pop. 2)



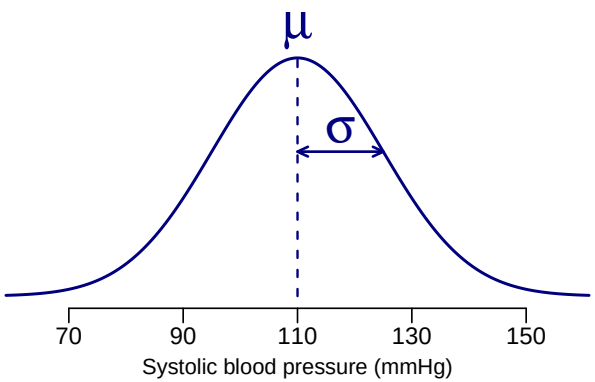
The underlying population gives rise to the data

3 **Data** $(x_1, \ldots, x_m)$ —$d(x_1, \ldots, x_m)$→ 4 $D = \{$ TRUE, FALSE $\}$

A decision function $d(x_1, \ldots, x_m)$ is a function whose domain is the set of possible values of the data and whose range is the set of possible decisions (decision space D)

A hypothesis about μ

1 **Population**: UK adults, $\mathcal{P}_{\mu,\sigma}$



The underlying population gives rise to the data

2 A **hypothesis** (H) is a statement about $\mathcal{P}_{\mu,\sigma}$

H: $\mu = 110$ mmHg
(i.e. the mean systolic blood pressure of UK adults is 110 mmHg)

3 **Data**: $(132, 128, 139, 125, 141, 134)$

$d(x_1, \ldots, x_m)$

4 $D = \{$ TRUE, FALSE $\}$

The decision function $d(x_1, \ldots, x_m)$ is **Student's *t*-test** at significance level α (e.g. 0.05).
It maps the observed data to $D = \{\text{TRUE, FALSE}\}$

Null hypothesis (H_0) and statistical errors

- We can never prove that a hypothesis (H_0) is TRUE
- We can **reject** H_0 (declare it FALSE), accepting a **controlled probability** α of rejecting it when it is actually TRUE

Hence, we want to control the probability of false positive statements (i.e. avoiding false statements about the population)

H_0	Not Rejected	Rejected
TRUE	$1-\alpha$	α
FALSE	β	$1-\beta$

$\alpha = \mathrm{P}(\text{reject } H_0 \mid H_0 \text{ TRUE})$: **Type I error rate** (false positive, significance level)

$\beta = \mathrm{P}(\text{fail to reject } H_0 \mid H_0 \text{ FALSE})$: **Type II error rate** (false negative)

$1-\beta = \mathrm{P}(\text{reject } H_0 \mid H_0 \text{ FALSE})$: **Statistical power** (true positive)

Hypothesis tests as decision functions

1. A hypothesis (H_0) makes a statement about a population (i.e. statistical model)



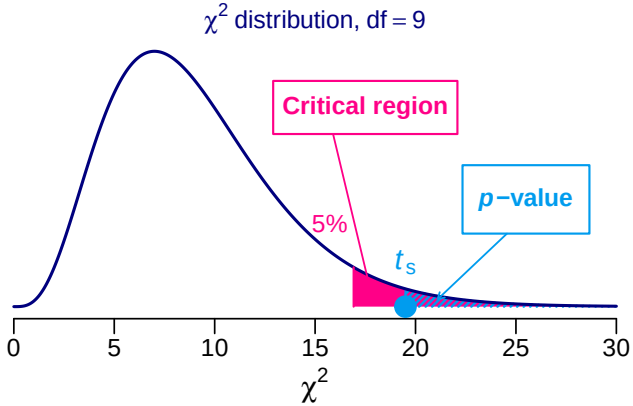
2. To test H_0, a sample from the population is collected



3. Data $(x_1, \ldots, x_m)$ are collected. A **test statistic t_s**, which is a function of the sample data, is calculated

$$t_s = t(x_1, \ldots, x_m)$$

4. Under H_0, the distribution of the test statistic t is known. The calculated test statistic t_s is located on this distribution



$d(t_s) =$ FALSE if t_s falls in the critical region

Student's *t*-test

1 H_0: the mean systolic blood pressure of UK adults is $\mu_0 = 110$ mmHg



2 To test H_0, a sample from the population is collected

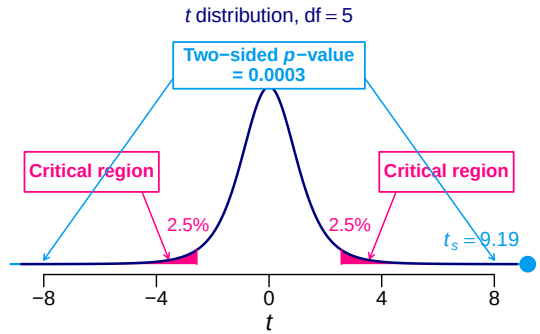


3 Data **(132, 128, 139, 125, 141, 134)** are collected. A test statistic t_s **(Student's *t*)**, which is a function of the sample data, is calculated

$$t_s = \frac{\bar{x} - \mu_0}{s/\sqrt{m}} = \frac{133.17 - 110}{6.18/\sqrt{6}}$$

$$= \frac{23.17}{2.52} = \mathbf{9.19}, \quad \text{df} = m - 1 = 5$$

4 Under H_0, the distribution of the test statistic t is known. The calculated test statistic t_s is located on this distribution



$d(t_s) =$ FALSE because t_s falls in the critical region

The multiple testing problem

Single null hypothesis

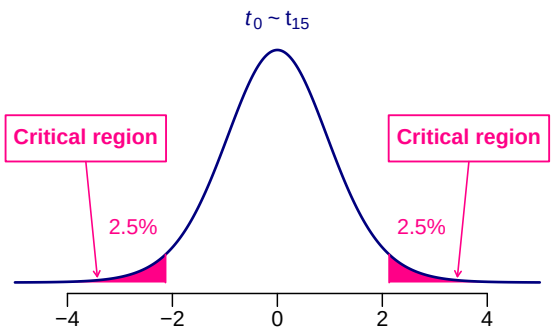
Let H_0 be a null hypothesis

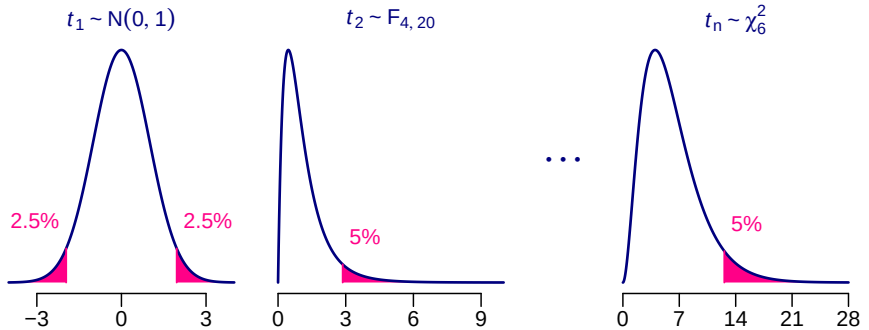
Family of null hypotheses

Let $\mathcal{F} = \{H_1, H_2, \dots, H_n\}$ be a family of null hypotheses

We apply a statistical test to H_0 and to each $H_i \in \mathcal{F}$

Under each null hypothesis, the distribution of its test statistic is known:





If H_0 is TRUE, the probability of a false positive statement is 5%

If all the H_i, $i = 1, 2, \dots, n$, are TRUE, the probability of at least one false positive statement (FWER) is $\geq 5\%$ and depends on the joint distribution of the test statistics t_i

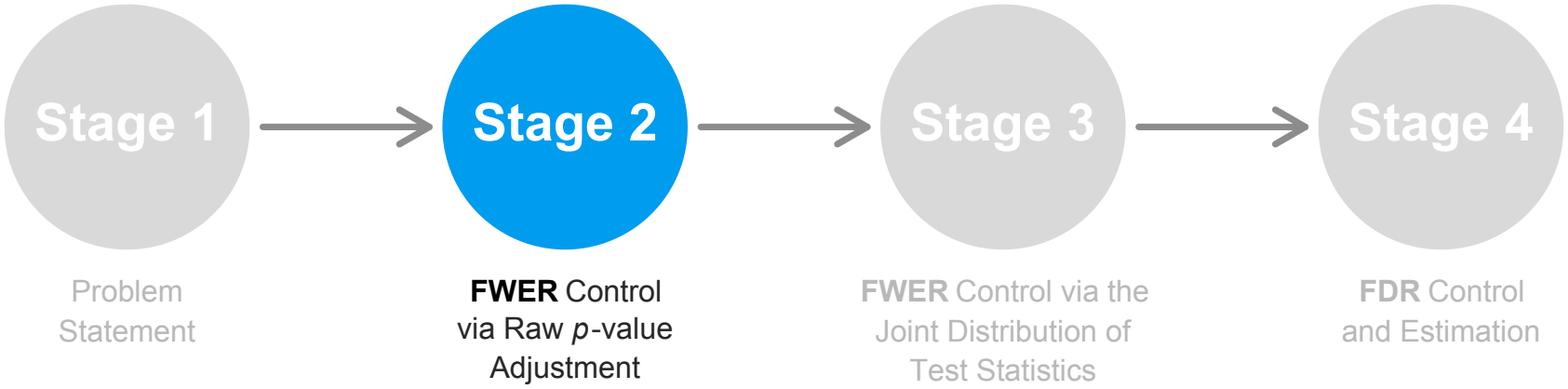
The multiple testing problem



If we go on testing long enough, we will inevitably find something that is "statistically significant"

A *p*-value controls the error rate of *its own* test, not that of the whole family

Roadmap



FWER: Family-Wise Error Rate

FDR: False Discovery Rate

Family-Wise Error Rate (FWER)

Let $\mathcal{F} = \{H_1, H_2, \ldots, H_n\}$ be a finite family of null hypotheses, and let m_0 ($0 \leq m_0 \leq n$) denote the number of TRUE **null hypotheses**.

$\mathcal{F}$	Not Rejected	Rejected	Total
TRUE	U	V	m_0
FALSE	T	S	$n - m_0$
Total	$n - R$	R	n

- Because the test statistics follow a probability distribution, the counts U, V, T, S, $n - R$ and R also follow probability distributions (i.e. they are random variables). The quantities $n - R$ and R are observable
- The counts m_0, $n - m_0$ and n are **fixed**, not random: n is known, whereas m_0 and $n - m_0$ are **unknown**

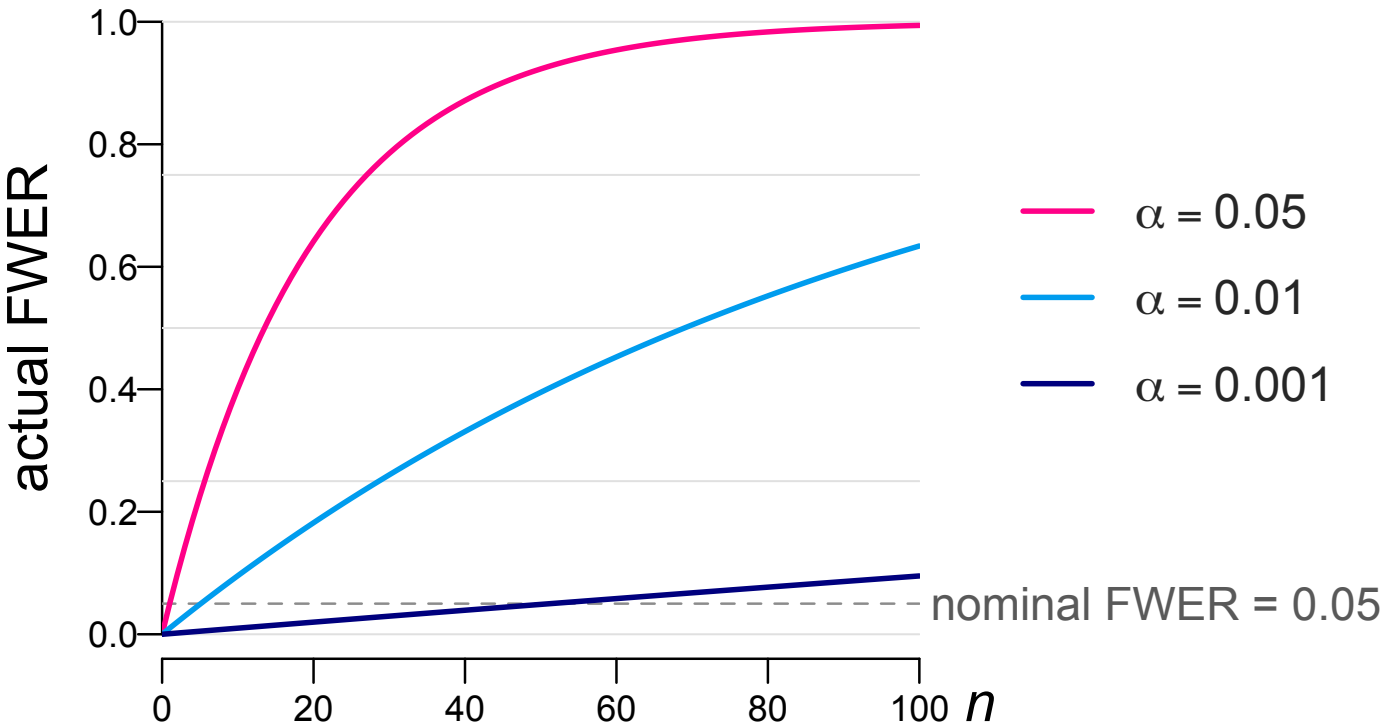
FWER $:=$ **P(** V **> 0)**

We want to control the FWER **whatever m_0 is** and whichever nulls are TRUE.

We accept only a small probability α (e.g. 0.05) of at least one false positive!!!

FWER and raw *p*-values

- Let $\mathcal{F} = \{H_1, H_2, \ldots, H_n\}$ be a finite family of TRUE null hypotheses: this is the **global null hypothesis**, H_0^{global}
- Each hypothesis is tested at a pre-specified significance level α
- Assuming the *n* tests are **independent**, FWER = $1 - (1 - \alpha)^n$



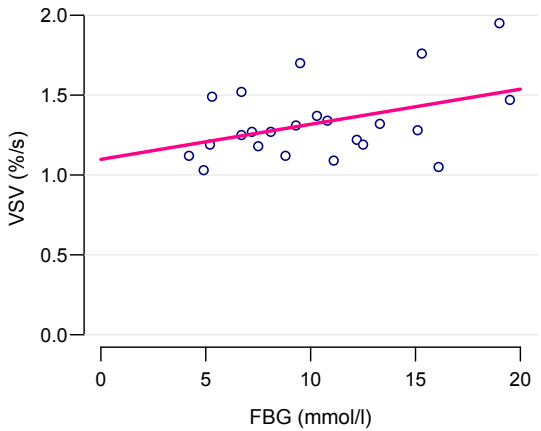
FWER and raw *p*-values

Ventricular shortening velocity (VSV, %/s) and fasting blood glucose (FBG, mmol/l) were measured in 24 type 1 diabetic patients (23 with complete data). The following two null hypotheses are tested:

H_1: $\mu_{VSV} = 0$ at FBG = 0

H_2: no linear association between FBG and VSV (i.e. slope = 0)

A simple linear model is used to test both hypotheses

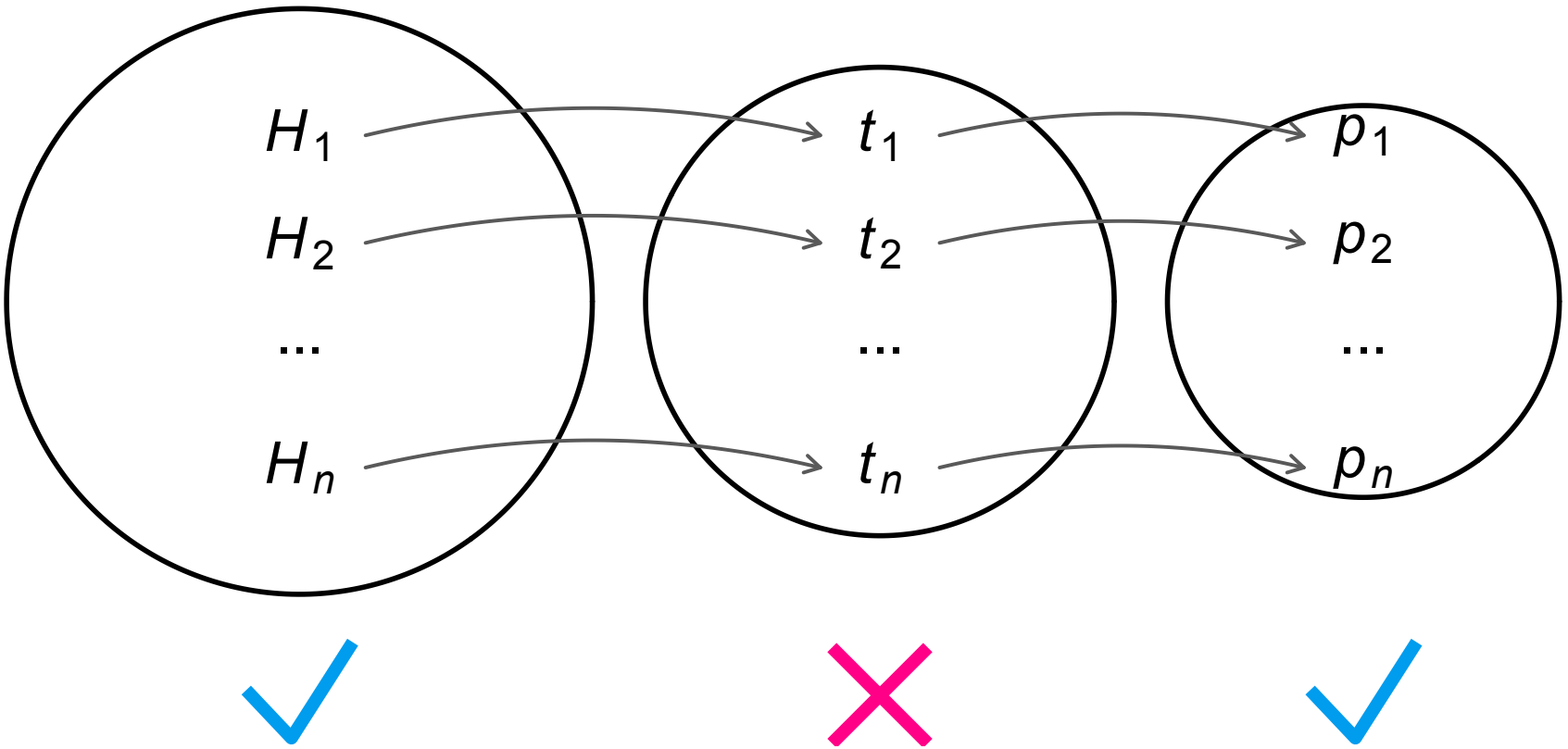



```
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.098    0.117      9.345   <0.001
## blood.glucose 0.022    0.010      2.101   0.048
```

Is this a false positive result?

Data: `thuesen` dataset (`ISwR` package v2.0-12, R 4.5.1)

Adjustment of raw *p*-values



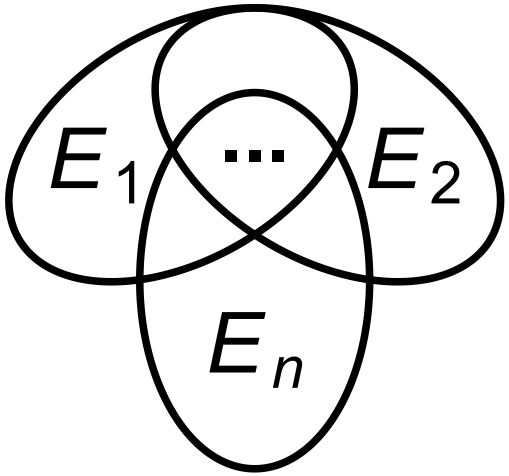
Here we do not use the multivariate distribution of the test statistics $(t_1, t_2, \ldots, t_n)$, working instead with the marginal *p*-values (p_i) of the individual null hypotheses (H_i).

Bonferroni correction

Boole's inequality

Let $E_1, E_2, \ldots, E_n$ be events.

$$\mathrm{P}\left(\bigcup_{i=1}^{n} E_i\right) \leq \sum_{i=1}^{n} \mathrm{P}(E_i)$$



Boole's inequality is applied in hypothesis testing

E_i := {Under a TRUE H_i, the test statistic t_i falls inside the **critical region**}

$$\text{FWER} = \mathrm{P}\left(\bigcup_{i \in K_0} E_i\right)$$ (K_0 = set of the m_0 TRUE null hypotheses)

- We want FWER $\leq \alpha$ (e.g. 0.05)
- Each test is run at level $\frac{\alpha}{n}$, so $\mathrm{P}(E_i) = \frac{\alpha}{n}$ for every $i \in K_0$
- Hence, FWER $\leq \sum_{i \in K_0} \mathrm{P}(E_i) = \frac{\alpha}{n} \times m_0 \leq \alpha$

Bonferroni correction

Ventricular shortening velocity (VSV, %/s) and fasting blood glucose (FBG, mmol/l) were measured in 24 type 1 diabetic patients (23 with complete data). The following two null hypotheses are tested:

H_1: $\mu_{\text{VSV}} = 0$ at FBG = 0

H_2: no linear association between FBG and VSV (i.e. slope = 0)

```
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.098    0.117      9.345   <0.001
## blood.glucose 0.022    0.010      2.101   0.048
```

Adjusted *p*-value

<0.001

0.096

Bonferroni does not reject H_2 at $\alpha = 0.05$

Data: `thuesen` dataset (`ISwR` package v2.0-12, R 4.5.1)

Correction rule

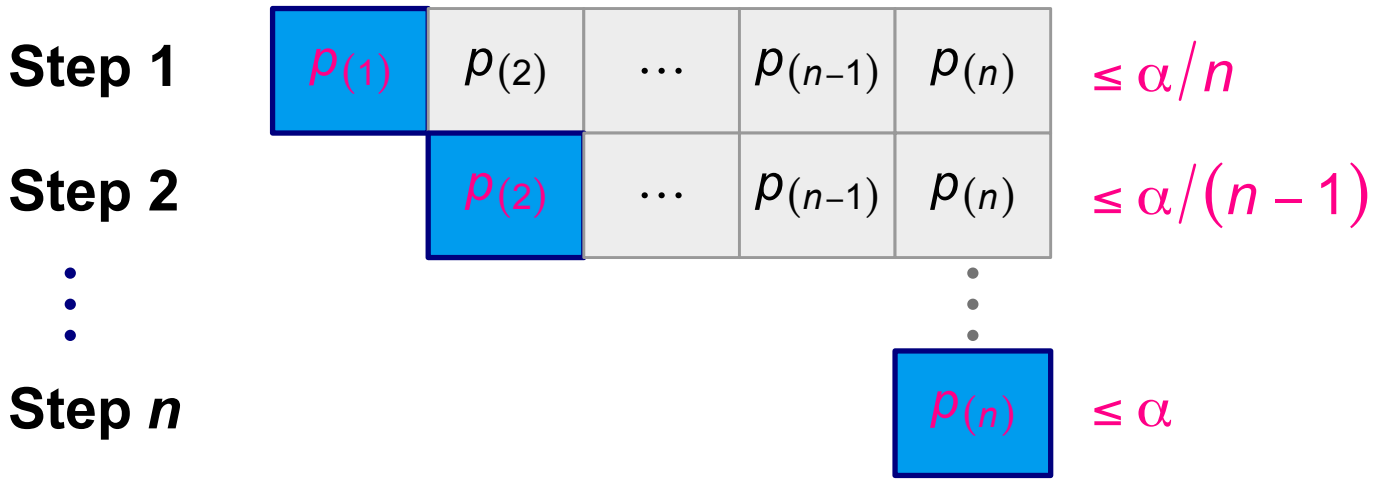
Raw *p*-values	**Adjusted *p*-values**
p_1	$n \times p_1$
p_2	$n \times p_2$
$\vdots$	$\vdots$
p_n	$n \times p_n$

Note: adjusted *p*-values must satisfy $\tilde{p}_i \leq 1$, hence $\tilde{p}_i = \min(1,\ n \times p_i)$

Holm–Bonferroni correction

We apply Boole's inequality in a step-down procedure

- Order the n p-values: $p_{(1)} \leq p_{(2)} \leq \ldots \leq p_{(n)}$, with corresponding hypotheses $H_{(1)}, H_{(2)}, \ldots, H_{(n)}$



- Stop at the first non-rejection: the remaining hypotheses are all retained

It controls the FWER at the α level, but it is less conservative than Bonferroni because the threshold **grows** after each rejection ($\frac{\alpha}{n}, \frac{\alpha}{n-1}, \frac{\alpha}{n-2}, \ldots$), whereas Bonferroni uses $\frac{\alpha}{n}$ throughout.

Holm–Bonferroni correction

Ventricular shortening velocity (VSV, %/s) and fasting blood glucose (FBG, mmol/l) were measured in 24 type 1 diabetic patients (23 with complete data). The following two null hypotheses are tested:

H_1: $\mu_{\text{VSV}} = 0$ at FBG = 0

H_2: no linear association between FBG and VSV (i.e. slope = 0)

```
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.098    0.117      9.345   <0.001
## blood.glucose 0.022    0.010      2.101   0.048
```

Adjusted *p*-value

<0.001

0.048

Holm rejects H_2 at $\alpha = 0.05$

Data: `thuesen` dataset (`ISwR` package v2.0-12, R 4.5.1)

Correction rule

Order the n p-values: $p_{(1)} \leq p_{(2)} \leq \ldots \leq p_{(n)}$

Raw *p*-values	**Adjusted *p*-values**
$p_{(1)}$	$n \times p_{(1)}$
$p_{(2)}$	$(n-1) \times p_{(2)}$
$\vdots$	$\vdots$
$p_{(n)}$	$1 \times p_{(n)}$

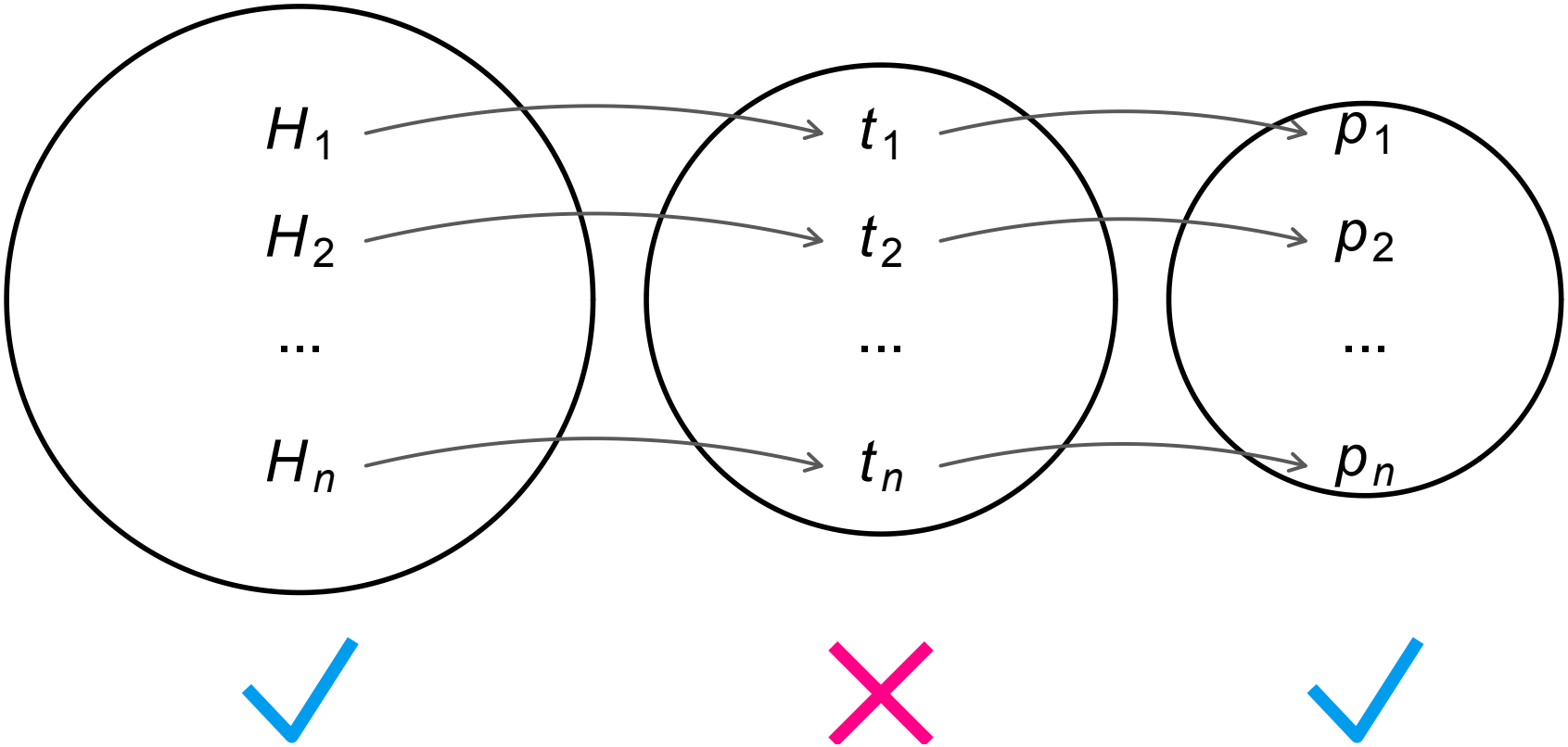
Note: adjusted p-values must satisfy

$$\tilde{p}_{(i)} \leq 1 \text{ and } \tilde{p}_{(i)} \leq \tilde{p}_{(i+1)}$$

General rule:

$$\tilde{p}_{(i)} = \max\left(\tilde{p}_{(i-1)},\ \min\left(1,\ (n-i+1) \times p_{(i)}\right)\right)$$

Assumptions on test statistic relationships



The corrections differ in what they assume about the **dependence** among the test statistics ($t_1, t_2, \ldots, t_n$): Bonferroni and Holm require **no assumption** about it, while Šidák assumes **independence**.

Šidák correction

- Let $H_1, H_2, \ldots, H_n$ be hypotheses with **independent** test statistics $t_1, t_2, \ldots, t_n$
- For each H_i, $i = 1, 2, \ldots, n$, set the significance level to $\alpha_{\text{Šidák}}$. Under H_0^{global},

$$\text{FWER} = 1 - (1 - \alpha_{\text{Šidák}})^n$$

- Set the FWER equal to α. Solving the previous equation for $\alpha_{\text{Šidák}}$, we obtain the individual significance level for each hypothesis H_i:

$$\alpha_{\text{Šidák}} = \mathbf{1} - (\mathbf{1} - \alpha)^{\frac{\mathbf{1}}{n}}$$

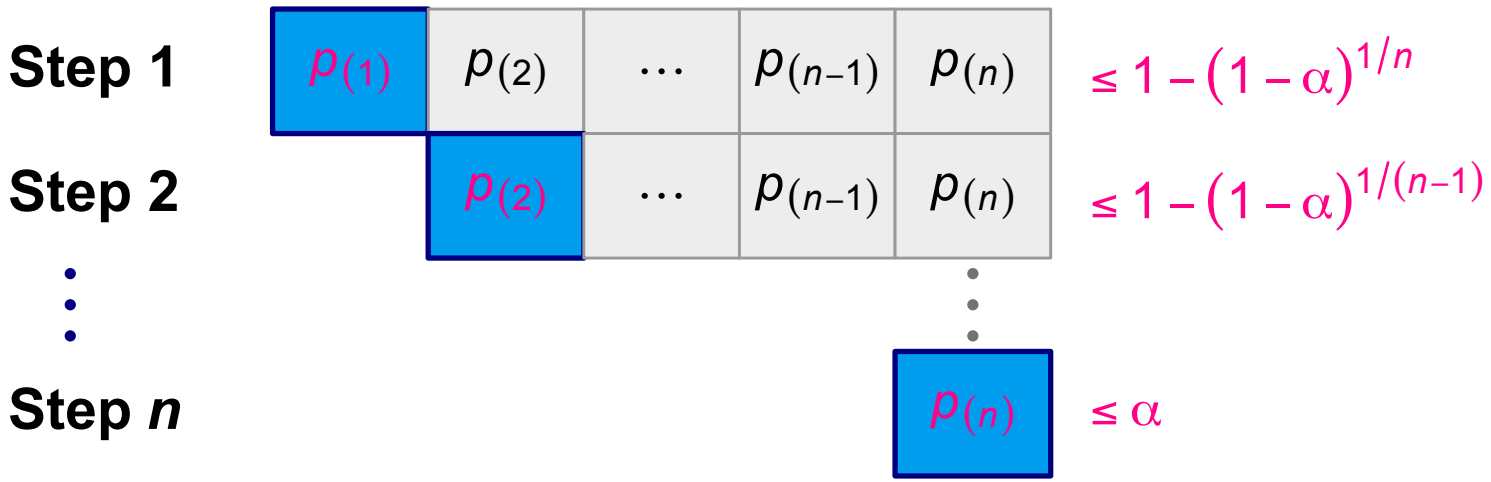
- Reject H_i if $p_i \leq \alpha_{\text{Šidák}}$, or equivalently if the adjusted p-value $\tilde{p}_i = 1 - (1 - p_i)^n \leq \alpha$

The Šidák correction is valid and, since $1 - (1 - \alpha)^{1/n} > \alpha/n$ for $n \geq 2$, it is slightly more powerful than the Bonferroni correction

Holm–Šidák correction

We apply the Šidák correction in a step-down procedure

- Order the n p-values: $p_{(1)} \leq p_{(2)} \leq \ldots \leq p_{(n)}$, with corresponding hypotheses $H_{(1)}, H_{(2)}, \ldots, H_{(n)}$ (**independent** test statistics)

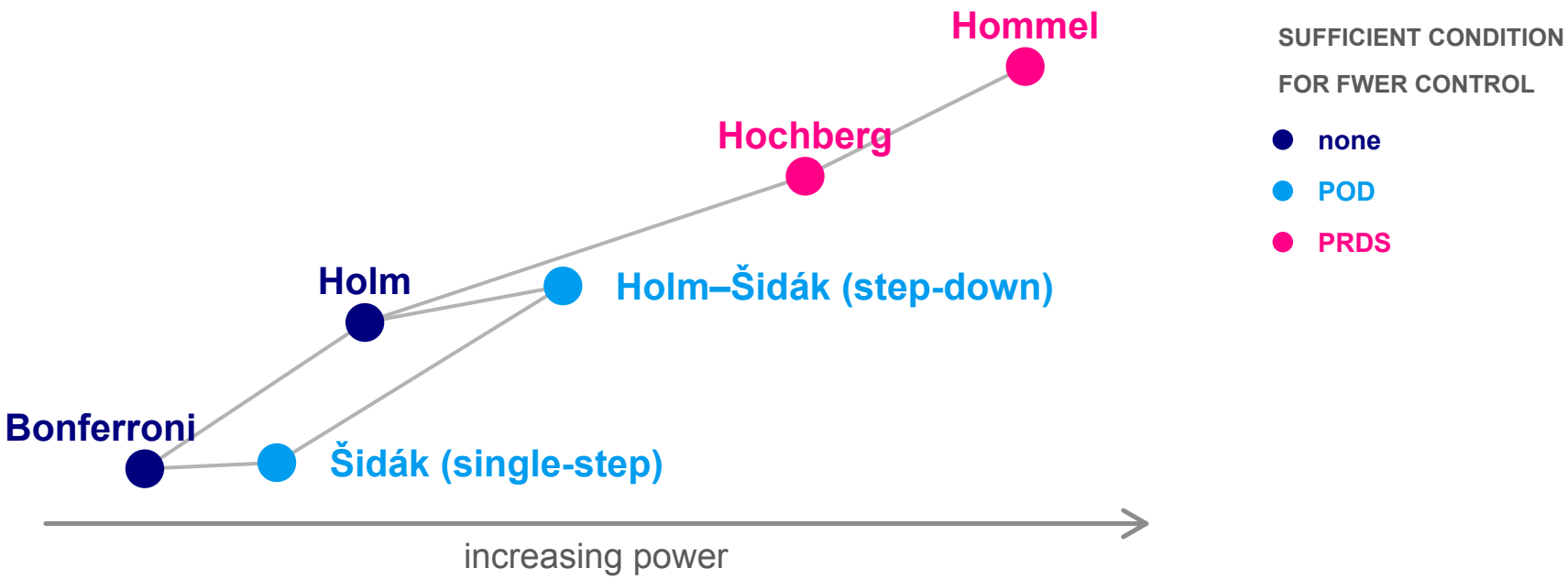


- Stop at the first non-rejection: the remaining hypotheses are all retained

It controls the FWER at the α level under independence, and it is slightly less conservative than Holm–Bonferroni since $1-(1-\alpha)^{1/j} \geq \alpha/j,\ j = 1, 2, \ldots, n$

Which correction is the most powerful?

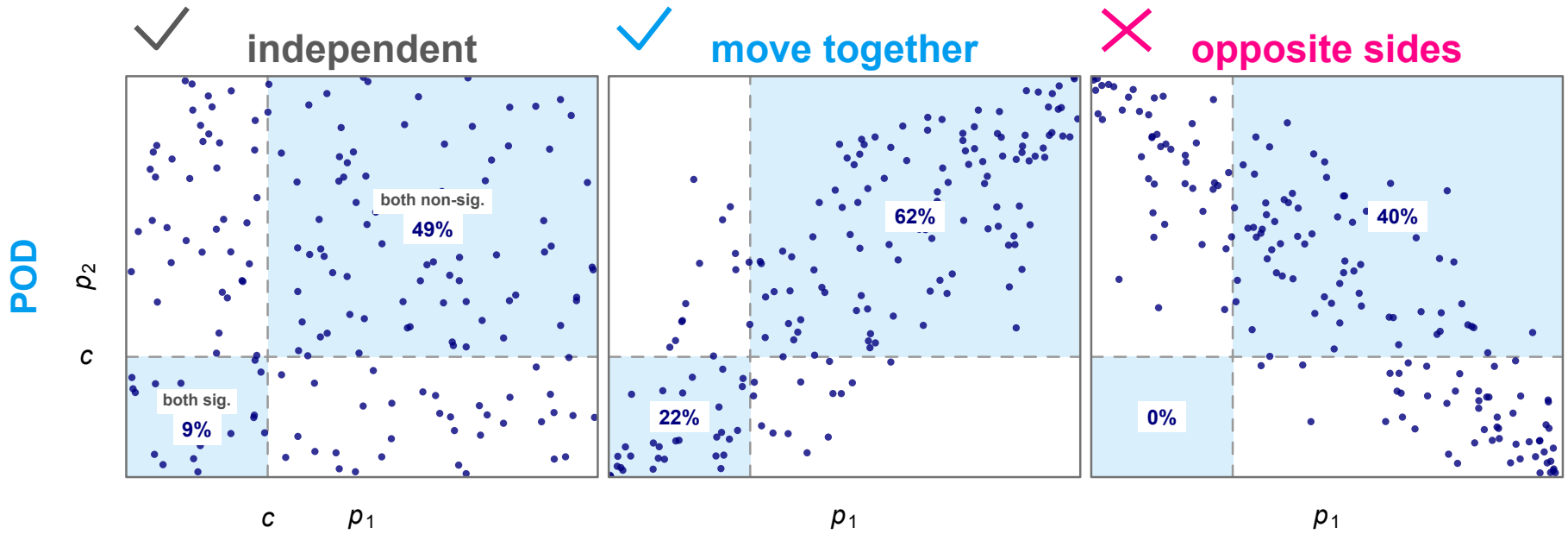
All six procedures **control** the FWER at α, each under its own condition. They differ in **how much power** they gain over Bonferroni, and in **what they must assume** to do so.



- **Šidák (single-step)** is **not uniformly more or less powerful** than **Holm**, **Hochberg** and **Hommel**: each can reject a hypothesis the other does not
- **Šidák**, **Holm–Šidák**, **Hochberg** and **Hommel** all **control the FWER** when the test statistics are **independent**

POD condition

POD = Positive Orthant Dependence

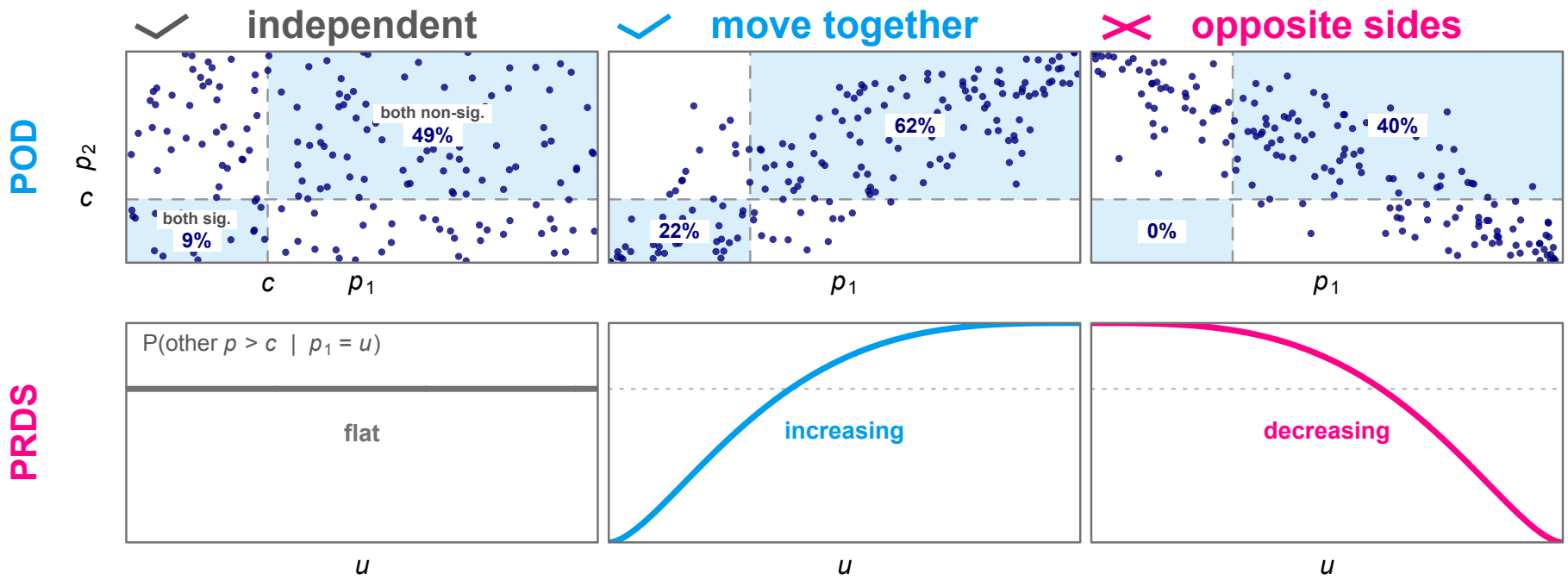


c = the **significance level** each p-value is compared with (e.g. 0.05)

- **POD**: for **any** threshold c (e.g. 0.05), the TRUE null p-values are **at least as likely** to be **all significant** (i.e. **at the same time**), **and** at least as likely to be **all non-significant** (i.e. **at the same time**), **as if they were independent**: $P(\text{all } p_i \leq c) \geq \prod_i P(p_i \leq c)$ and $P(\text{all } p_i > c) \geq \prod_i P(p_i > c)$. **Šidák** needs **only** the second one (positive upper orthant dependence, PUOD): $\text{FWER} = 1 - P(\text{all } p_i > c) \leq 1 - (1-c)^{m_0} \leq \alpha$ **(all together, not one given another)**

PRDS condition

PRDS = Positive Regression Dependence on a Subset



u = a **possible value taken** by the p-value of a TRUE null H_i, $u \in [0, 1]$

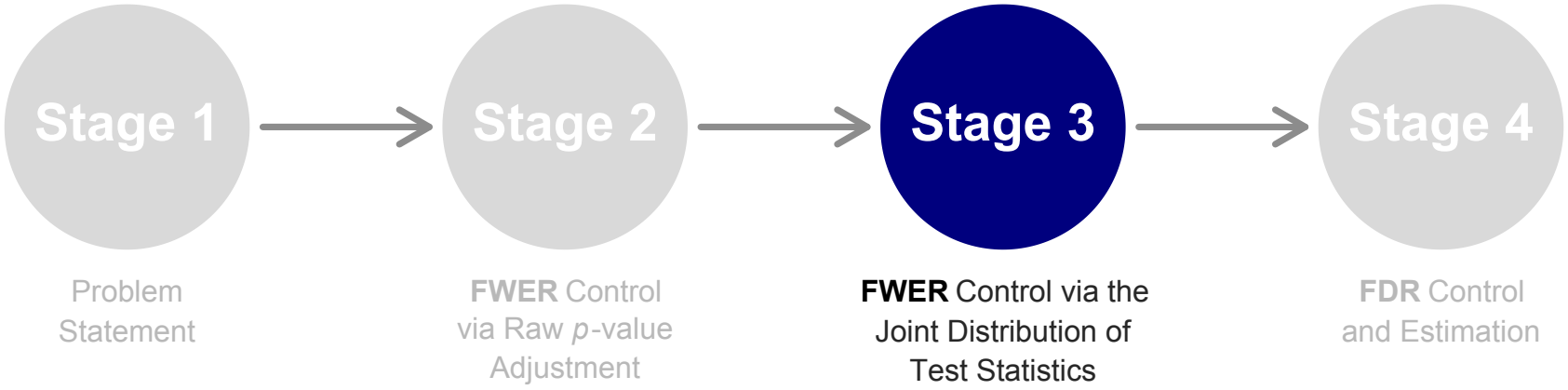
- **PRDS**: **POD** compares **one unconditional joint probability** with the **independent** case, while **PRDS** requires that, **for each** TRUE **null** H_i, $i \in K_0$, the probability that the **other** p-values fall **outside the critical region** be **non-decreasing in** u, **at every** u. **Sufficient** for **Hochberg** and **Hommel**, and for **BH** to control the **FDR**
- **PRDS implies POD**, but **not** the converse

Simulations and practicals

Shiny web applications

https://bioinformatics.cruk.cam.ac.uk/stats/shinystats/

Roadmap



FWER: Family-Wise Error Rate

FDR: False Discovery Rate

Joint distribution of the test statistics

Ventricular shortening velocity (VSV, %/s) and fasting blood glucose (FBG, mmol/l) were measured in 24 type 1 diabetic patients (23 with complete data). The following two null hypotheses are tested:

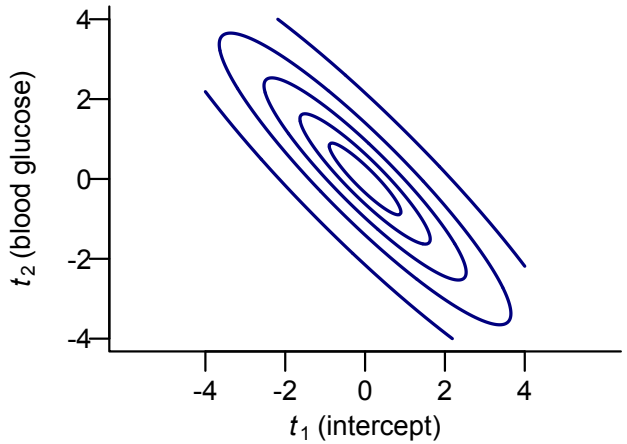
H_1: $\mu_{\text{VSV}} = 0$ at FBG $= 0$

H_2: no linear association between FBG and VSV (i.e. slope $= 0$)

```
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.098    0.117      9.345   <0.001
## blood.glucose 0.022    0.010      2.101   0.048
```

Data: `thuesen` dataset (`ISwR` package v2.0-12, R 4.5.1)

Under H_0^{global}, we know the joint distribution of the test statistics



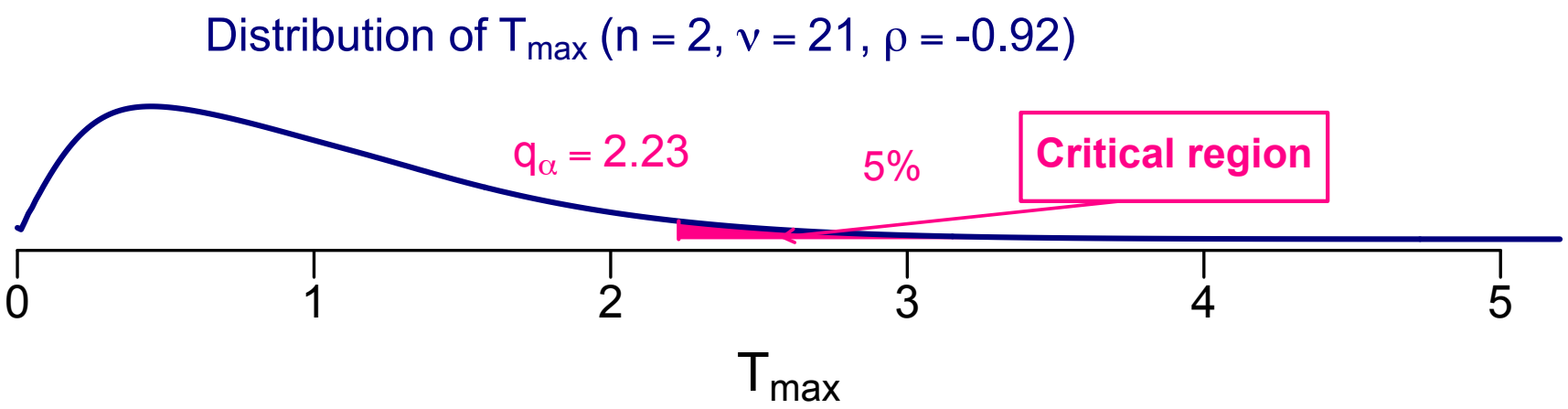
Bivariate Student's t distribution under H_0^{global} ($\nu = 21$, $\rho = -0.92$)

The max-*t* statistical principle

- Let $\mathcal{F} = \{H_1, H_2, \ldots, H_n\}$ be a finite family of null hypotheses
- Let $\mathcal{D}$ be the joint distribution of the individual test statistics $t_1, t_2, \ldots, t_n$ under H_0^{global} (i.e. all hypotheses are TRUE). As in our example, the tests are two-sided: larger values of $|t_i|$ favour the rejection of H_i
- Consider the maximum of the individual test statistics t_i:

$$T_{\max} = \max_{i=1,2,\ldots,n} |t_i| \qquad (\text{one-sided tests: } T_{\max} = \max_i t_i)$$

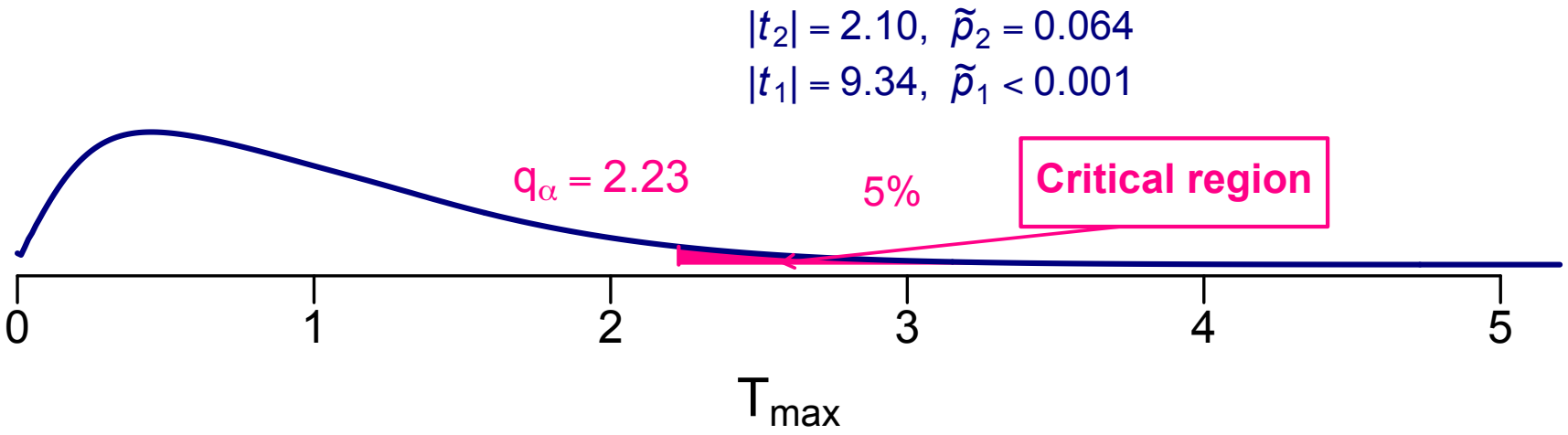
The joint distribution $\mathcal{D}$ determines the distribution of $T_{\max}$ under H_0^{global}:



Data: `thuesen` dataset (`ISwR` package v2.0-12, R 4.5.1)

The max-*t* statistical principle

- Consider the positions of the statistics $|t_1|, |t_2|, \ldots, |t_n|$:



- H_1 is **rejected**: $|t_1| = 9.34 > q_\alpha = 2.23$, adjusted $\tilde{p}_1 < 0.001$
  H_2 is **not rejected**: $|t_2| = 2.10 < q_\alpha = 2.23$, adjusted $\tilde{p}_2 = 0.064 > 0.05$
  (single-step max-*t*: adjusted $\tilde{p}_2 = 0.064$; step-down max-*t*: adjusted $\tilde{p}_2 = 0.048$)

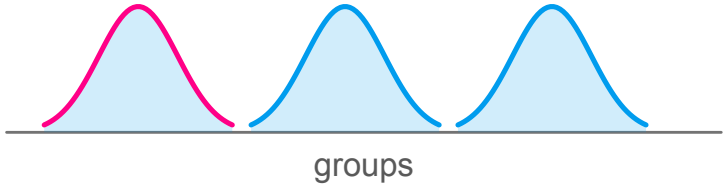
Subset pivotality condition: take **any** subset K of the hypotheses. Their statistics $\{t_i : i \in K\}$ must behave the same way when only the hypotheses in K are TRUE as they do when **all** of them are TRUE

Under the subset pivotality condition, the FWER is controlled (FWER $\leq \alpha$**)**

How to apply the max-*t* principle for multiple comparisons

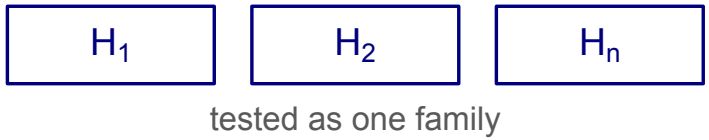
1 **Statistical model**

what the data look like, and what is assumed



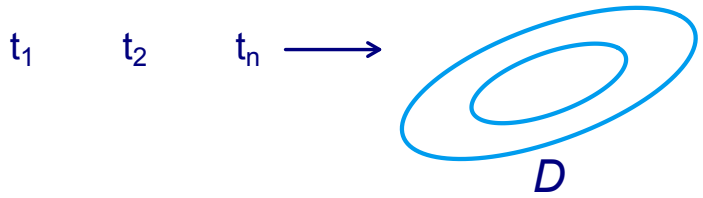
2 **Hypotheses**

one per comparison, all tested together



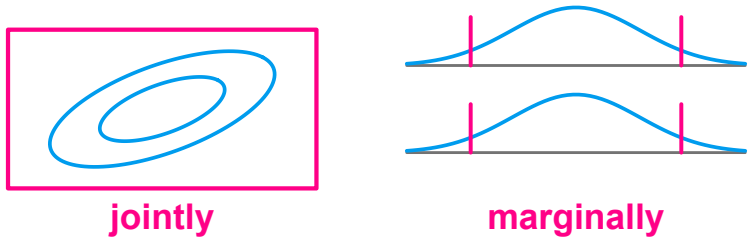
3 **Test statistics and their joint law**

each t_i, and D = their joint law under H_0^{global}



4 **How the joint law is used**

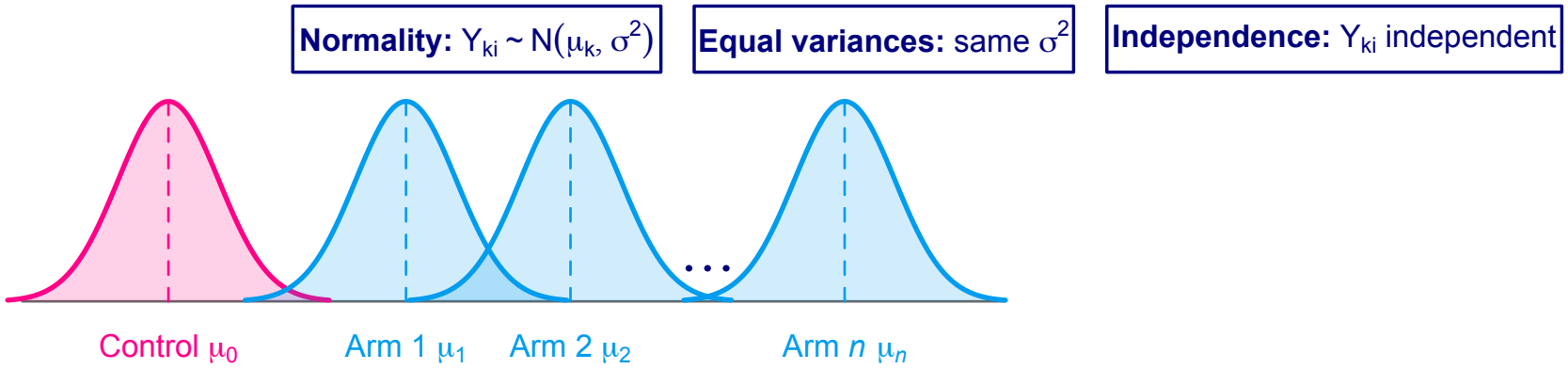
jointly or **marginally** (as if uncorrelated)



Jointly: critical value(s) from $\mathcal{D}$, correlations used. **Marginally**: each one alone, as if uncorrelated

Dunnett's test

- **Statistical model**: one-way analysis of variance (ANOVA)



- **Hypotheses**: $H_k : \mu_k = \mu_0, k = 1, \ldots, n$

  For each of the n arms, the arm mean equals the control mean μ_0

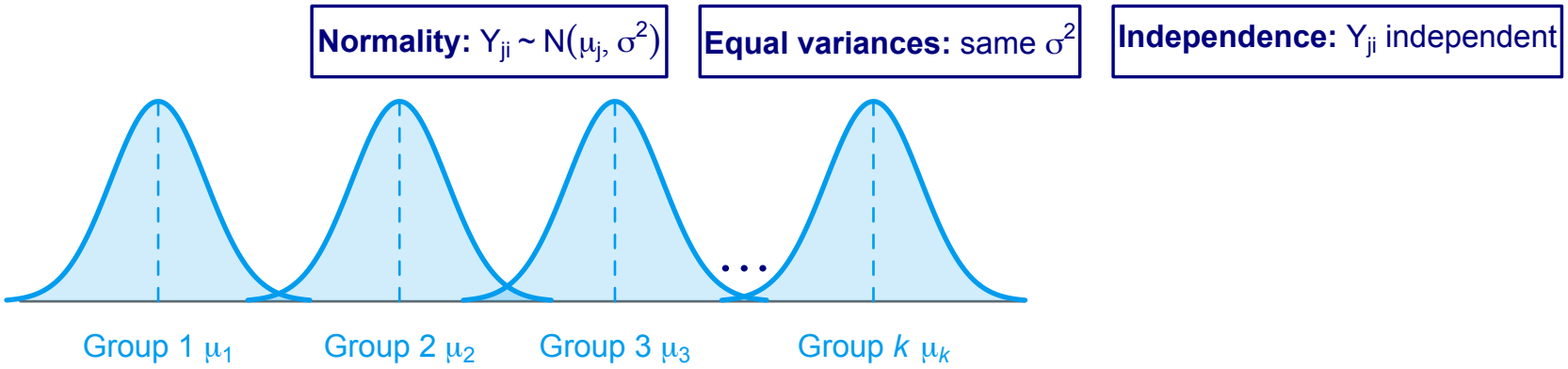
- Each t_k follows a Student's t distribution with $N - (n + 1)$ degrees of freedom under H_k (N = total sample size)

  $\mathcal{D}$ is a multivariate Student's t distribution under H_0^{global}

  → **The max-t principle is applied to $\mathcal{D}$**

Tukey's test

- **Statistical model**: one-way analysis of variance (ANOVA)



- **Hypotheses**: $H_{j\ell} : \mu_j = \mu_\ell, 1 \leq j < \ell \leq k$

  For each of the $n = k(k-1)/2$ pairs, the two group means are equal

- Each $t_{j\ell}$ follows a Student's t distribution with $N - k$ degrees of freedom under $H_{j\ell}$

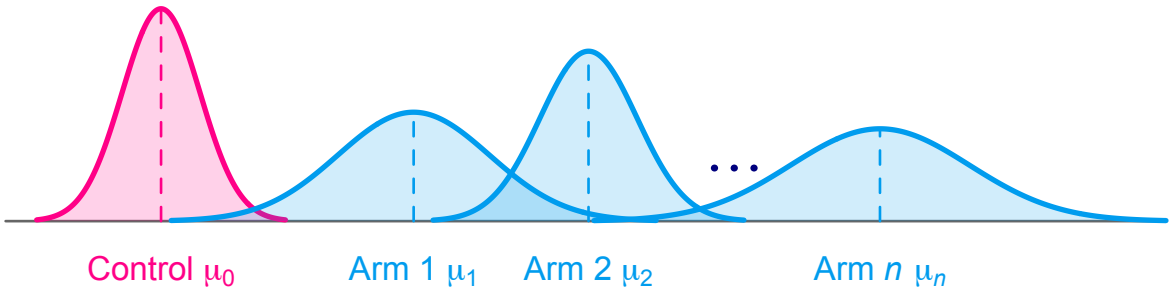
  $\mathcal{D}$ is a multivariate Student's t distribution under H_0^{global}

  → **The max-t principle is applied to $\mathcal{D}$**

Dunnett's test with unequal variances (Hasler–Hothorn)

- **Statistical model**: one-way ANOVA **without** the equal-variance assumption

Normality: $Y_{ki} \sim N(\mu_k, \sigma_k^2)$ | ~~**Equal variances:** same σ^2~~ | **Independence:** Y_{ki} independent



- **Hypotheses**: $H_k : \mu_k = \mu_0, k = 1, \ldots, n$

  For each of the n arms, the arm mean equals the control mean μ_0

- Each t_k (Welch) follows **approximately** a Student's t distribution under H_k

  $\mathcal{D}$ is **approximately** a multivariate Student's t under H_0^{global}

  (Satterthwaite df, **estimated** correlations; needs **moderate** n_k)

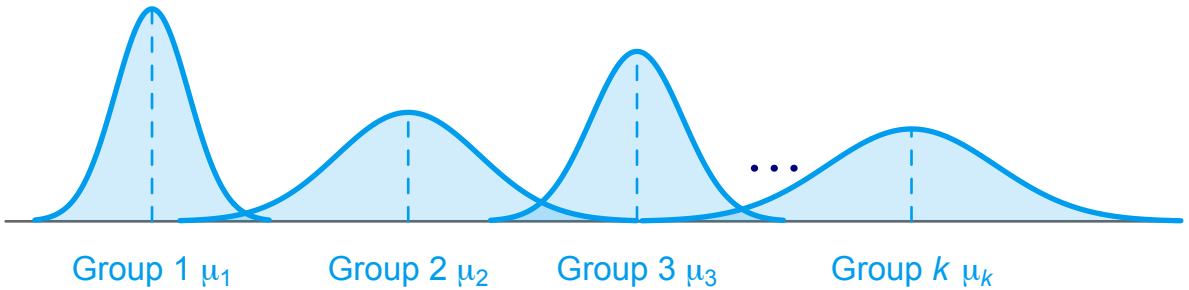
  ↳ **The max-t principle is applied to $\mathcal{D}$**

The **Šidák version** treats $\mathcal{D}$ **marginally**: each statistic on its own

Tukey's test with unequal variances (Hasler–Hothorn)

- **Statistical model**: one-way ANOVA **without** the equal-variance assumption

Normality: $Y_{ji} \sim N(\mu_j, \sigma_j^2)$ | ~~**Equal variances:** same σ^2~~ | **Independence:** Y_{ji} independent



- **Hypotheses**: $H_{j\ell} : \mu_j = \mu_\ell, 1 \le j < \ell \le k$

  For each of the $n = k(k-1)/2$ pairs, the two group means are equal

- Each $t_{j\ell}$ (Welch) follows **approximately** a Student's t distribution under $H_{j\ell}$

  $\mathcal{D}$ is **approximately** a multivariate Student's t under H_0^{global}

  (Satterthwaite df, **estimated** correlations; needs **moderate** n_k)

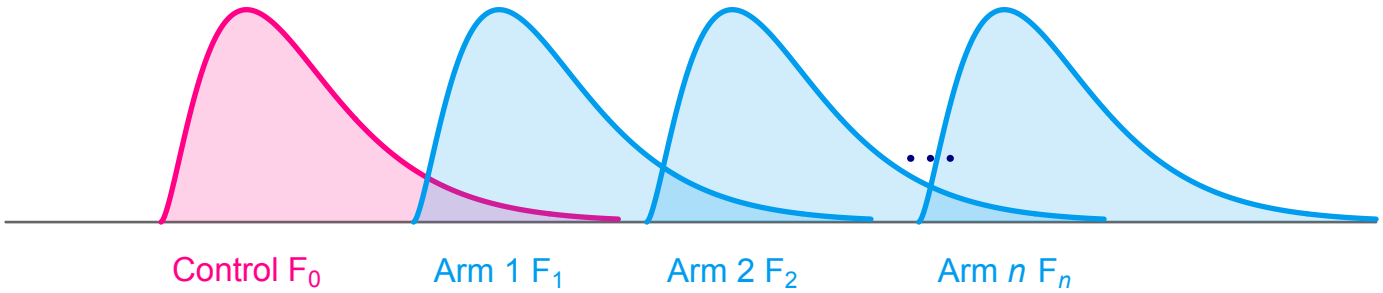
  ↳ **The max-t principle is applied to $\mathcal{D}$**

T2 and **T3** use $\mathcal{D}$ **marginally**; **Games–Howell** jointly: **Welch** statistics, **Tukey's** critical value

Dunnett-type test without normality (Steel)

- **Statistical model**: one-way layout **without** the normality assumption

~~**Normality:** $Y_{ki} \approx N(\mu_k, \sigma^2)$~~ | **Same distribution:** within each pair | **Independence:** Y_{ki} independent



- **Hypotheses**: $H_k : F_k = F_0, k = 1, \ldots, n$

  For each of the n arms, the arm distribution equals the control distribution F_0

- Each t_k (standardised Wilcoxon rank sum, **ranked within the control–arm pair only**) is **asymptotically** standard normal under H_k (no ties)

  $\mathcal{D}$ is **asymptotically** a multivariate normal distribution under H_0^{global}

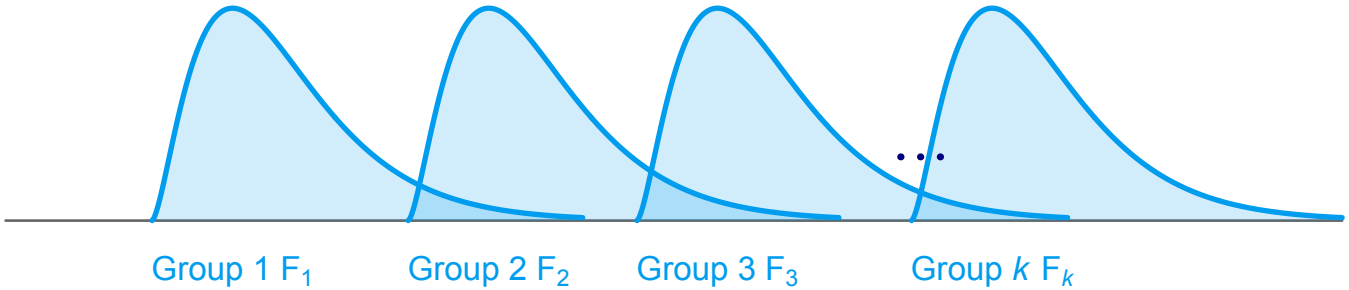
  → **The max-t principle is applied to $\mathcal{D}$**

The **Šidák version** treats $\mathcal{D}$ **marginally**: each statistic on its own

Tukey-type test without normality (Steel–Dwass)

- **Statistical model**: one-way layout **without** the normality assumption

~~**Normality:** $Y_{ji} \sim N(\mu_j, \sigma^2)$~~ | **Same distribution:** within each pair | **Independence:** Y_{ji} independent



- **Hypotheses**: $H_{j\ell} : F_j = F_\ell, 1 \le j < \ell \le k$

  For each of the $n = k(k-1)/2$ pairs, the two group distributions are equal

- Each $t_{j\ell}$ (standardised Wilcoxon rank sum, **ranked within each pair only**) is **asymptotically** standard normal under $H_{j\ell}$ (no ties)

  $\mathcal{D}$ is **asymptotically** a multivariate normal distribution under H_0^{global}

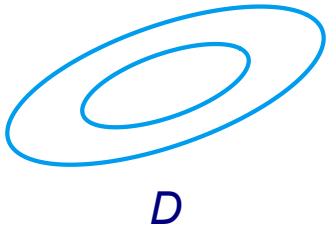
  → **The max-t principle is applied to $\mathcal{D}$**

The **Šidák version** treats $\mathcal{D}$ **marginally**: each statistic on its own

How to apply non-max-*t* procedures for multiple comparisons

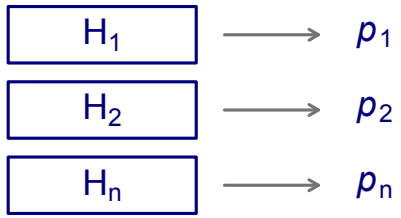
1 **Joint distribution**

same start as max-*t*: D, but only its margins



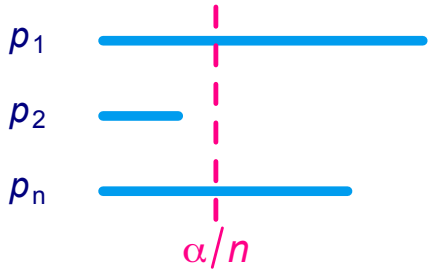
2 **Test by hypothesis**

one test each, from its own margin of D



3 **Correction**

Bonferroni or Holm on the raw *p*-values



One test per hypothesis from its **margin** of $\mathcal{D}$, then multiplicity by **correcting the *p*-values**

Dunn's test (not a max-*t* procedure)

- **Statistical model**: one-way layout **without** normality, all *k* groups **alike**

~~**Normality:** $Y_{ji} \sim N(\mu_j, \sigma^2)$~~ | **Same distribution:** all *k* groups | **Independence:** Y_{ji} independent



$F_1 = F_2 = F_3 = F_k$

- **Hypotheses**: $H_{j\ell} : F_j = F_\ell, 1 \leq j < \ell \leq k$

  For each of the $n = k(k-1)/2$ pairs, the two group distributions are equal

- **Test statistic** for each pair, from the **mean ranks** $\bar{R}_j$ over all *N* observations, standardised under H_0^{global} (no ties): a ***p*-value for each pair**

- **Bonferroni** or **Holm** is then applied to these *n* *p*-values: only the **margins** of $\mathcal{D}$ are used

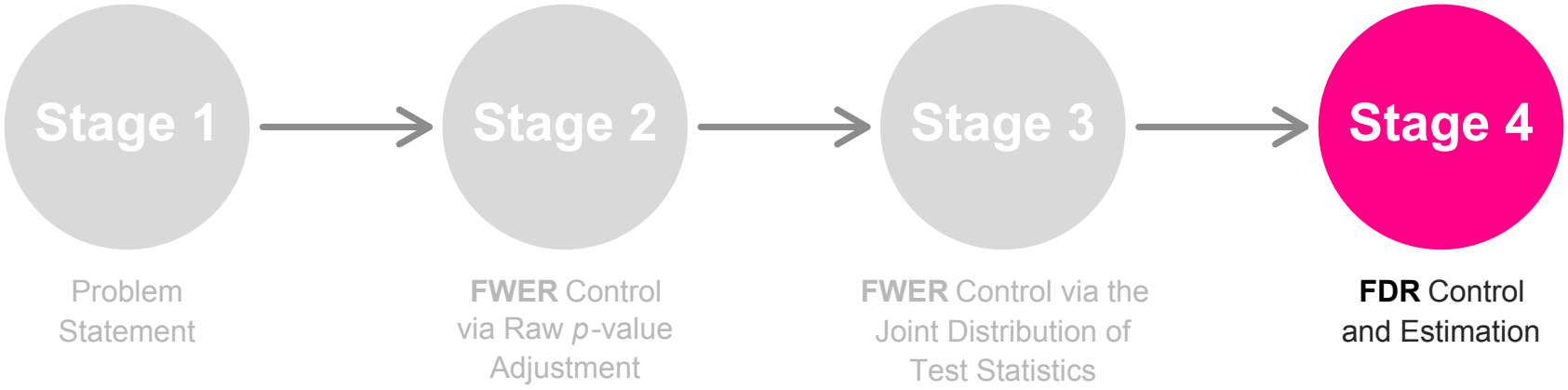
Dunn assumes **all *k* groups have the same distribution**; otherwise **FWER** can **far exceed** α

Simulations and practicals

Shiny web applications

https://bioinformatics.cruk.cam.ac.uk/stats/shinystats/

Roadmap

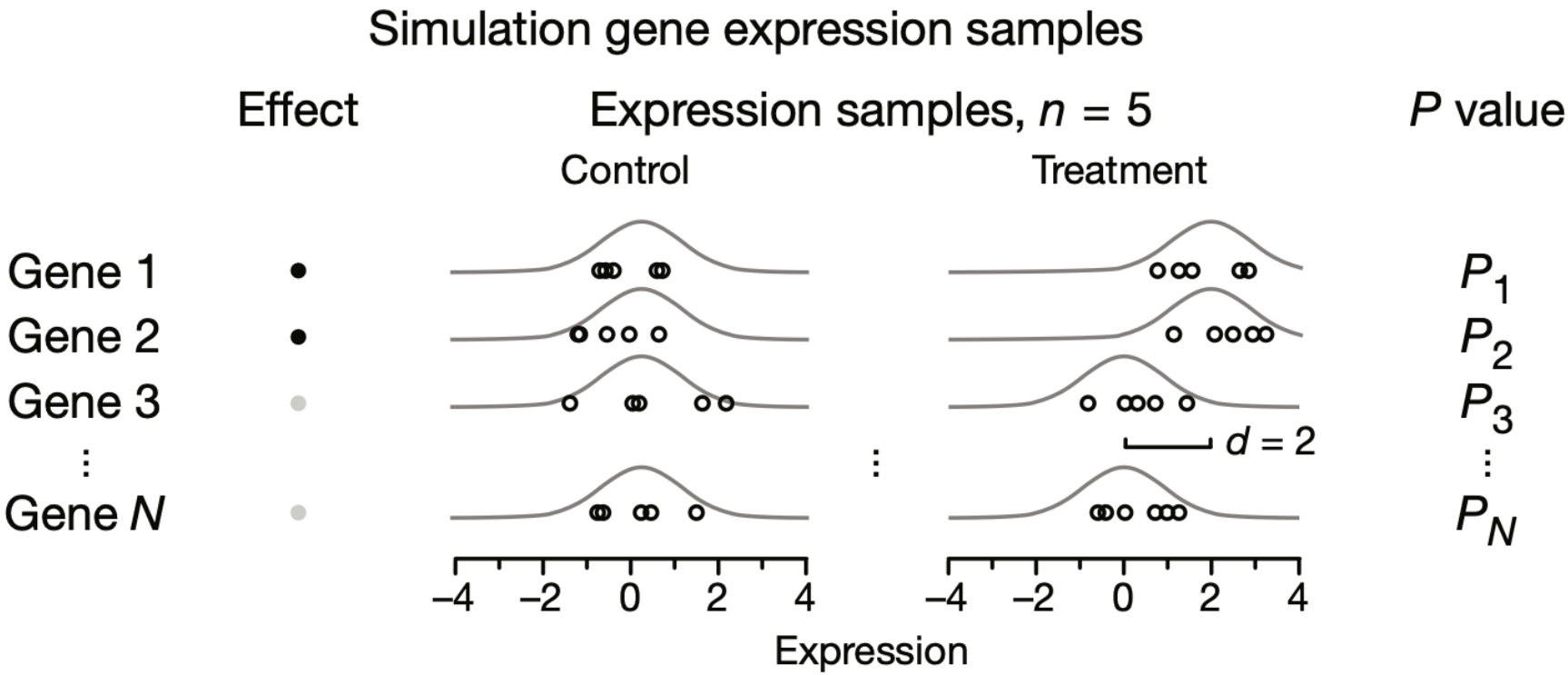


FWER: Family-Wise Error Rate

FDR: False Discovery Rate

FWER and statistical power: the Achilles' heel

For each gene, expression was simulated in a control and a treatment group (5 units each), drawn from a normal distribution ($\mu = 0$, $\sigma = 1$). For a fraction of genes (10% and 50%), an effect size ES $= 2$ ($\sim$ 79% power, $\alpha = 0.05$) was simulated by setting $\mu = 2$ in the treatment sample



Genes were assumed to be statistically independent.

Krzywinski, M. & Altman, N. *Comparing samples — part II*. Nature Methods **11**(4), 355–356 (April 2014).

FWER and statistical power: the Achilles' heel

% of truly differentially expressed genes declared significant

	10% affected			50% affected		
n	Bonf.	Holm	Šidák	Bonf.	Holm	Šidák
10	34.95	35.04	35.37	34.76	37.79	35.15
100	8.70	8.75	8.85	8.72	9.01	8.88
1,000	1.49	1.49	1.52	1.49	1.50	1.52
10,000	0.20	0.20	0.21	0.20	0.20	0.21
100,000	0.02	0.02	0.02	0.02	0.02	0.02

Legend: n = number of genes; Bonf. = Bonferroni correction; 10^5 simulated trials per n

- Power collapses as the family grows: with $n = 10$, about 35% of the truly affected genes are detected; with $n = 100{,}000$, only 0.02%
- All three corrections give essentially the same power. The step-wise gain (Holm) is visible only for small n with many true effects

False discovery rate (FDR)

Let $\mathcal{F} = \{H_1, H_2, \ldots, H_n\}$ be a finite family of null hypotheses, and let m_0 ($0 \leq m_0 \leq n$) denote the number of TRUE **null hypotheses**.

$\mathcal{F}$	Not Rejected	Rejected	Total
TRUE	U	V	m_0
FALSE	T	S	$n - m_0$
Total	$n - R$	R	n

- Consider the **proportion** of rejections that are false, $Q = V/R$, with the convention $Q = 0$ when $R = 0$
- Q is a random variable that cannot be observed: R is known once the tests are run, but V is not. We therefore control its **expectation**

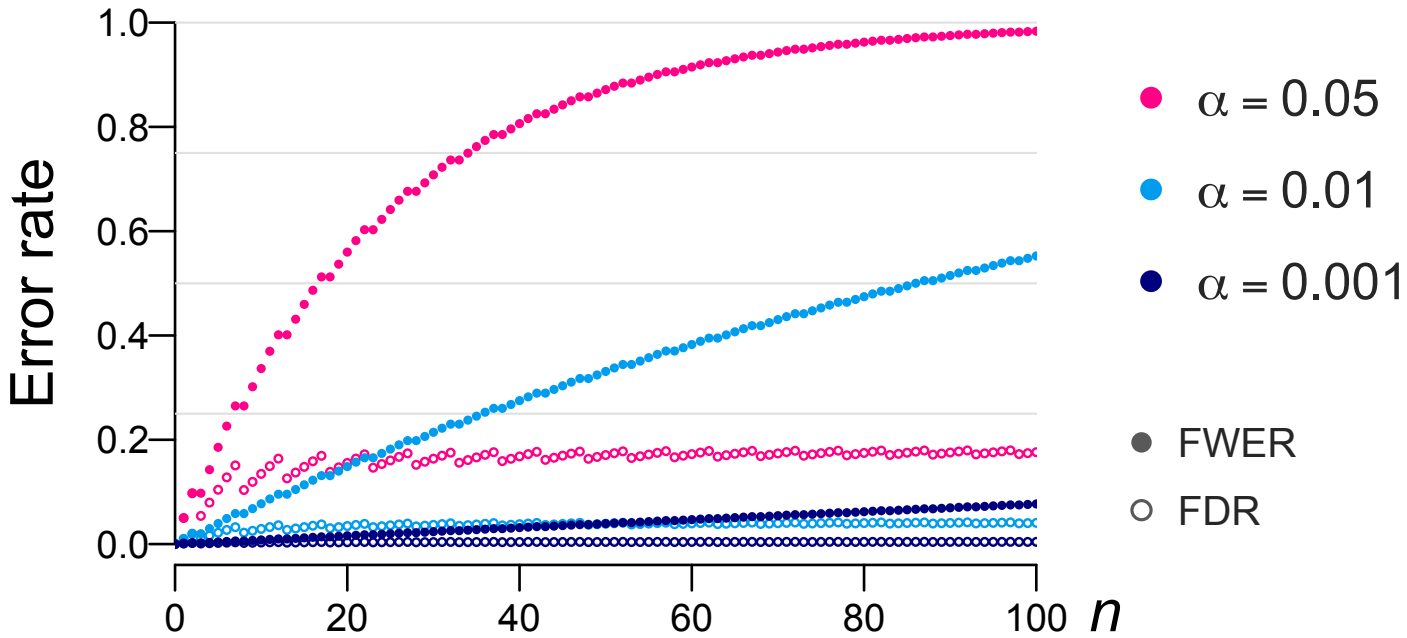
$$\mathbf{FDR} := \mathbf{E}\left[Q \right]$$

We accept that, **on average**, a controlled **proportion of our discoveries is false**.
We tolerate some errors in exchange for more power.

FWER and FDR as the number of tests grows

- Let $\mathcal{F} = \{H_1, H_2, \ldots, H_n\}$ be a finite family of null hypotheses with **independent** test statistics, each tested at level α
- In both series 80% of the nulls are TRUE ($m_0 = 0.8\,n$, no effect); the other 20% are FALSE, each detected with an assumed **power** = **90%** at every α

- **Filled**: FWER $= 1 - (1-\alpha)^{m_0}$ (a probability)
- **Open**: FDR $= \mathrm{E}[V/R]$ (an expected proportion)

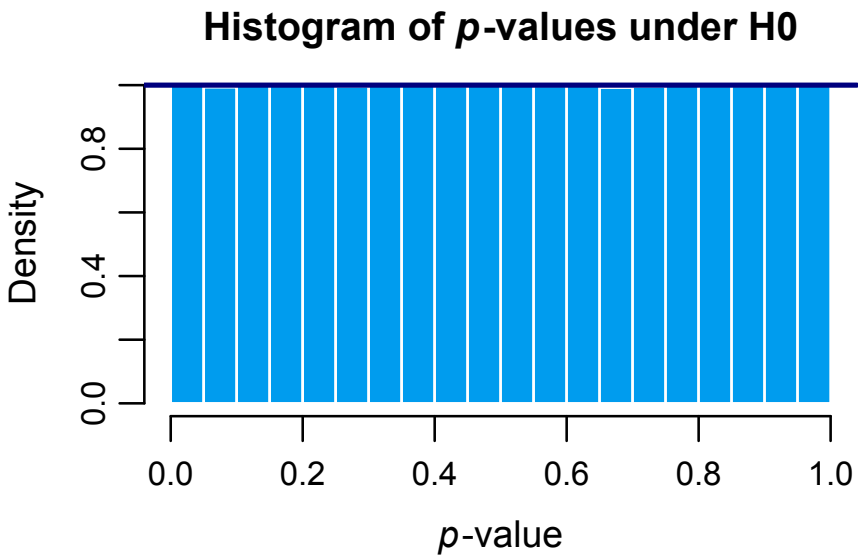


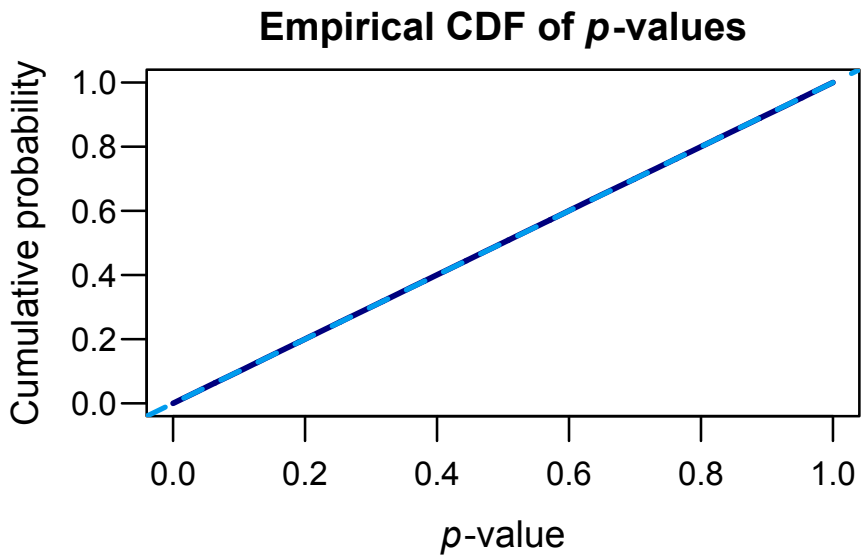
Three questions about the FDR

- **How can we control the FDR?**
- **How can we estimate the FDR?**
- **How can we compute FDR-adjusted *p*-values?**

Uniform distribution of the *p*-value

Let the null hypothesis H_0 be TRUE, and let the *p*-value be computed from the **continuous** null distribution of the **test statistic** *t*. The *p*-value (*p*) is itself a random variable such that $\boldsymbol{p \sim \mathrm{Uniform}(0, 1)}$.

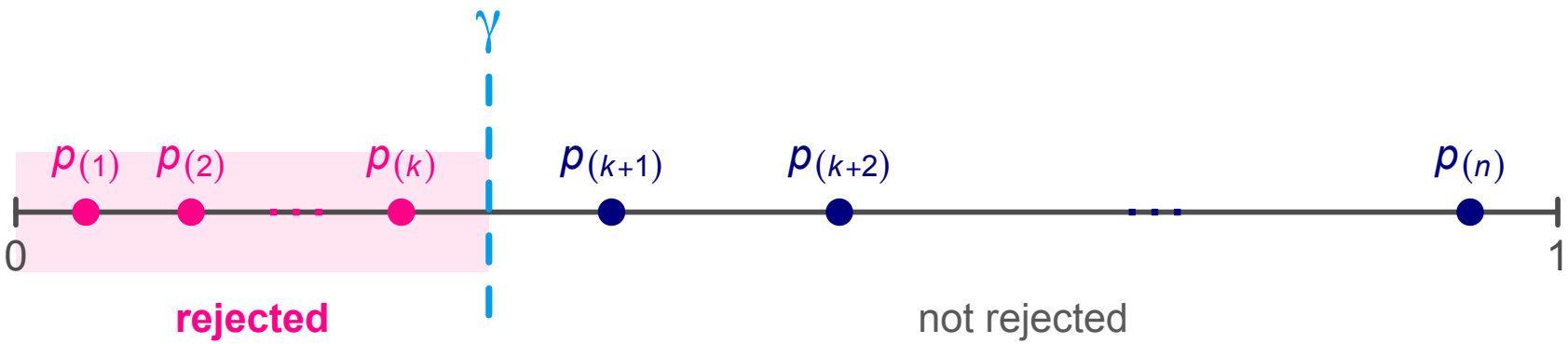




Simulated *p*-values under H_0 ($\mu_1 = \mu_2$) from a two-sample (pooled) Student *t*-test with 5 units per group (10^6 simulations): the histogram is flat at density 1 and the empirical cumulative distribution function (CDF) tracks the diagonal $y = x$, confirming $p \sim \mathrm{Uniform}(0, 1)$.

Controlling the FDR: Benjamini–Hochberg (BH)

- Let $p_{(1)} \leq p_{(2)} \leq \ldots \leq p_{(n)}$ be the ordered p-values: (i) is the **rank**



All p-values below γ are **rejected**

γ is chosen so that the **estimated** FDR below γ is at most q (e.g. $q = 0.10$)

- With k the number of p-values below γ, and $f(\gamma)$ the probability that a TRUE null p-value falls below γ:

$$\mathbb{E}(V) = f(\gamma)\, m_0 \leq f(\gamma)\, n \implies \text{FDR} \approx \frac{f(\gamma)\, n}{k} \implies \boldsymbol{f(\gamma) \leq \frac{k\, q}{n}}$$

Controlling the FDR: Benjamini–Hochberg (BH)

- γ is the **threshold**, and $f(\gamma)$ is the **probability** that a TRUE null p-value falls below γ
- From the **distribution of the p-value**: under H_0 it is Uniform$(0, 1)$, so the probability that it falls below γ is γ, whatever m_0 is

$$\text{Hence} \quad \boldsymbol{f(\gamma) = \gamma}$$

- Hence γ must satisfy the following inequality:

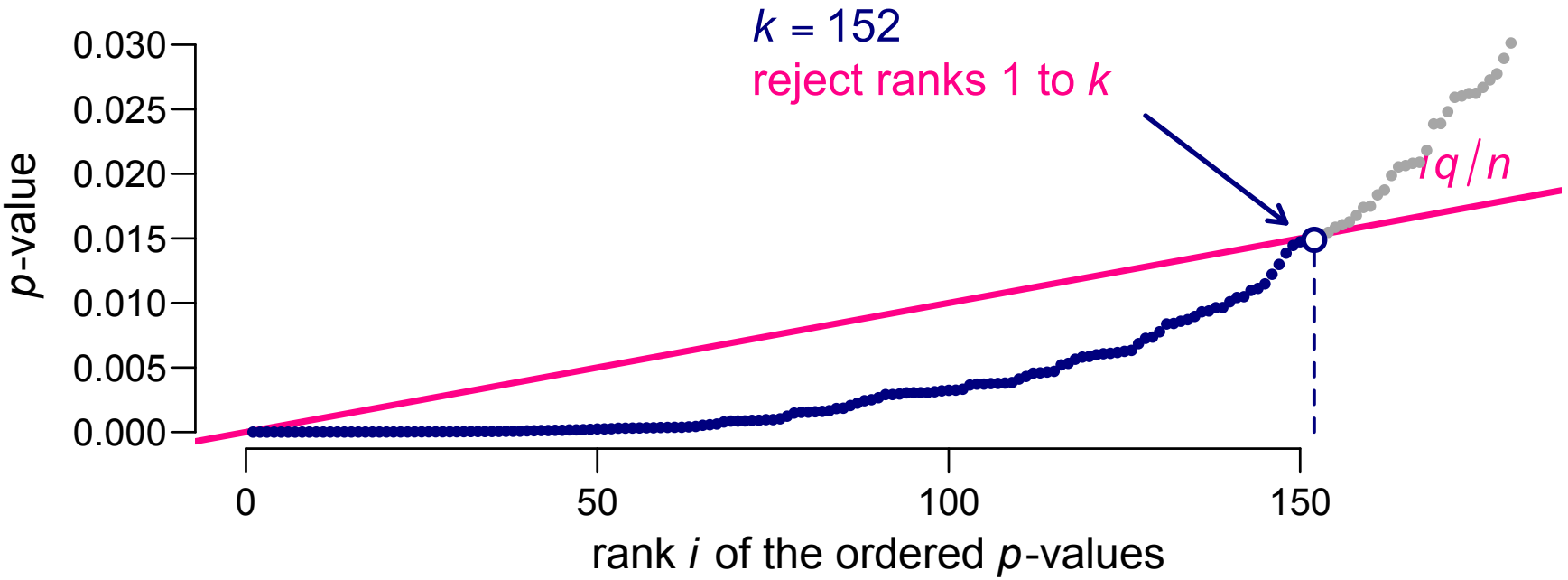
$$\gamma \leq \frac{\boldsymbol{k\,q}}{\boldsymbol{n}}$$

- **BH** chooses the **largest** γ that satisfies this inequality (k is the number of p-values below γ)

Benjamini, Y. & Hochberg, Y. *Controlling the false discovery rate: a practical and powerful approach to multiple testing*. J. R. Stat. Soc. B **57**(1), 289–300 (1995).

Controlling the FDR: Benjamini–Hochberg (BH)

The **ordered** p-values from $n = 1000$ **independent** tests against the threshold iq/n ($q = 0.10$): BH rejects up to the **largest** i with $p_{(i)} \leq iq/n$



rank i	$p_{(i)}$	iq/n	$\tilde{p}_{(i)}$
152 **(= *k*)**	0.014884	0.015200	**0.09792**
153 **(= *k+1*)**	0.015304	0.015300	0.10003

The BH-adjusted p-value $\tilde{p}_{(i)}$: see the slide *FDR-adjusted p-values*

Controlling the FDR: Benjamini–Hochberg (BH)

BH controls the FDR at level q when the p-values are **independent**, or have a specific **positive dependence** among the TRUE nulls (PRDS)

drop the assumption

estimate $\pi_0 = m_0/n$

BY

no assumption on the dependence

at the price of power

Storey

more powerful

needs independence (or only local correlation, with many tests)

Benjamini, Y. & Yekutieli, D. *The control of the false discovery rate in multiple testing under dependency*. Ann. Statist. **29**(4), 1165–1188 (2001).

Storey, J. D., Taylor, J. E. & Siegmund, D. *Strong control, conservative point estimation and simultaneous conservative consistency of false discovery rates*. J. R. Stat. Soc. B **66**(1), 187–205 (2004).

Controlling the FDR: Benjamini–Yekutieli (BY)

BH and BY are the **same** step-up rule — order the p-values, take the **largest** k such that

$$p_{(k)} \le \frac{k\,q}{n\,c(n)}$$

BH: $c(n) = 1$

BY: $c(n) = \sum_{j=1}^{n} \frac{1}{j} \approx \ln n + 0.577$ (large n)

- Assumptions: **none** on the dependence of the p-values
- The price: $c(n) > 1$ makes the threshold $c(n)$ times **stricter at every rank** — not $c(n)$ times fewer discoveries ($c(100) \approx 5.2$, $c(1000) \approx 7.5$, $c(10^4) \approx 9.8$)
- In our $n = 1000$ example, at $q = 0.10$: BH rejects 152 hypotheses, BY only **75**

BY is the dependence-robust version of BH

Controlling the FDR: Storey's procedure

- BH is **calibrated** for the worst case $m_0 = n$: every hypothesis is a TRUE null. Storey instead takes into account the proportion of true nulls $\pi_0 = m_0/n$
- π_0 is unknown, but it can be **estimated from the data**: $\hat{\pi}_0$
- Apply the **BH rule** with $q/\hat{\pi}_0$ in place of q: take the **largest** k such that

$$p_{(k)} \leq \frac{k\,q}{n\,\hat{\pi}_0}$$ and reject the k smallest p-values

- Since $\hat{\pi}_0 \leq 1$ (it is capped at 1), the threshold is **at least as large** as BH's: **more power**, and the more hypotheses are false ($\pi_0 \ll 1$), the larger the gain
- Assumptions: independence, or only **local** correlation, and **many** tests for a reliable $\hat{\pi}_0$

How $\hat{\pi}_0$ is estimated: see the slides *Storey's estimate of the FDR*

Controlling the FDR: which method?

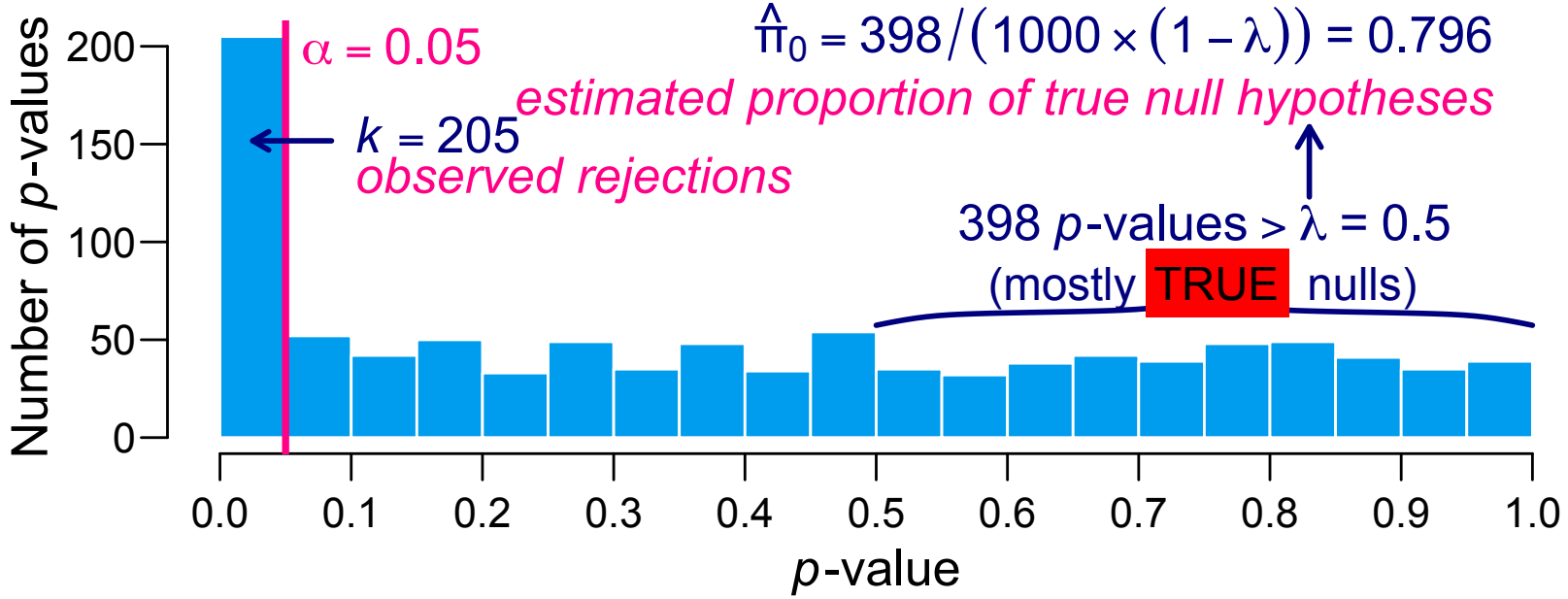
Method	Assumptions	Power	Best used for
BH	Independence or PRDS	Intermediate	Positively correlated tests; few true effects ($\pi_0 \approx 1$); too few tests to estimate π_0
BY	Any dependence	Lowest	Unknown or negative dependence
Storey	Independence or local correlation, plus a reliable $\hat{\pi}_0$	Highest	Many tests, large fraction of true effects ($\pi_0 \ll 1$)

All three order the *p*-values and use a step-up threshold: they differ only in the constant *dividing* $k\,q/n$: fixed (BH, BY) or *estimated* (Storey)

	BH	BY	Storey
Threshold $\gamma \leq$	$\frac{k\,q}{n}$	$\frac{k\,q}{n\,c(n)}$	$\frac{k\,q}{n\,\hat{\pi}_0}$

Storey's estimate of the FDR

The same $n = 1000$ **independent** tests: empirical distribution of the p-values, with $\alpha = 0.05$ used as a **fixed cut-off** and $\lambda = 0.5$ above which p-values are taken as TRUE nulls



$\hat{\pi}_0$ is capped at 1: $\hat{\pi}_0 = \min\{1,\ W/(n(1-\lambda))\}$

$$\widehat{\mathrm{FDR}}(\alpha) = \frac{\hat{\pi}_0 \times n \times \alpha}{k} = \frac{0.796 \times 1000 \times 0.05}{205} = \mathbf{0.194}$$

Storey, J. D. *A direct approach to false discovery rates*. J. R. Stat. Soc. B **64**(3), 479–498 (2002).

Storey's estimate of the FDR

- **Storey errs on the safe side**: it counts **every** *p*-value above 0.5 as a TRUE null, but a few real effects land there too. So it **overestimates** how many nulls are TRUE, and its estimated FDR is slightly **too high**, never too low

- **Local correlation**: genes are correlated only within **small clusters** of a few co-regulated genes, positive **or** negative, with the clusters unrelated to each other

- **Global correlation must be avoided**: Storey counts the *p*-values of **this one study**. Small clusters each go their own way and **average out**, so the count is reliable. But if **all** the genes move together they are **one** big cluster: the whole picture shifts from study to study, and the count can be badly wrong — even when every correlation is **positive**

FDR-adjusted *p*-values: the *q*-value

- **Definition**: the **smallest FDR** at which a test would still be called significant.
  Note: $q_{(i)}$ is the ***q*-value of one test**, not the FDR **level** q

- **Calculation of *q*-values**: for the ordered *p*-values $p_{(1)} \leq \cdots \leq p_{(n)}$

BH	**BY**	**Storey**
$\frac{n}{j} p_{(j)}$	$c(n) \frac{n}{j} p_{(j)}$	$\hat{\pi}_0 \frac{n}{j} p_{(j)}$

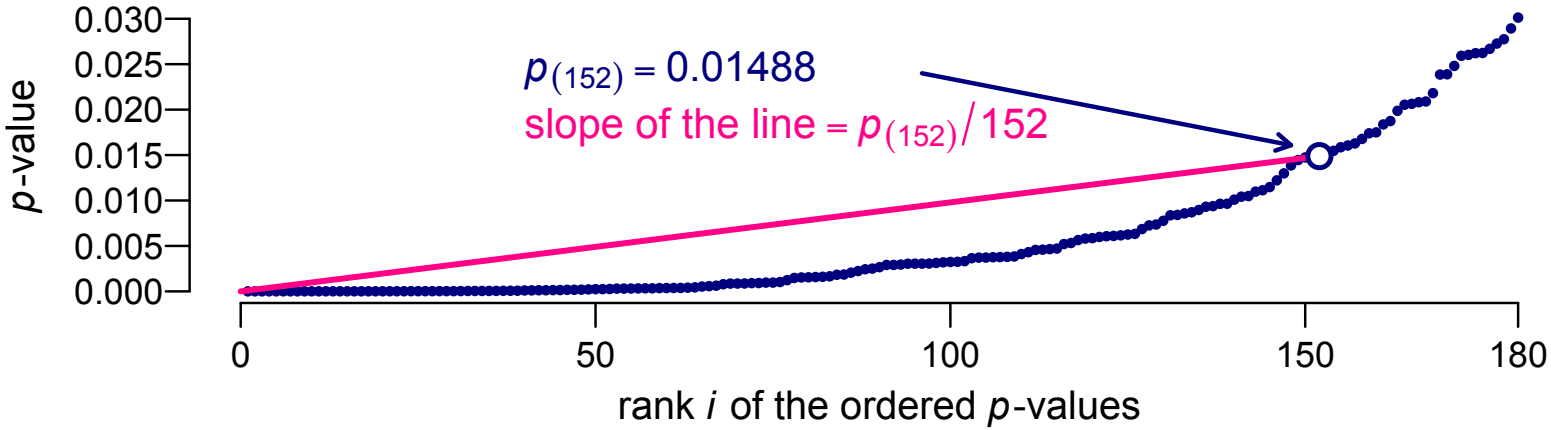
Two constraints: a *q*-value can never be above **1**, and a larger *p*-value must never give a smaller *q*-value. Both are met by taking $q_{(i)}$ = the **smallest** of these values from rank *i* **onwards**

Only the constant in front differs: 1 (BH), $c(n) = \sum_{j=1}^{n} 1/j$ (BY), $\hat{\pi}_0$ (Storey)

Storey, J. D. *The positive false discovery rate: a Bayesian interpretation and the q-value*. Ann. Statist. **31**(6), 2013–2035 (2003). Storey, J. D. & Tibshirani, R. *Statistical significance for genomewide studies*. Proc. Natl. Acad. Sci. USA **100**(16), 9440–9445 (2003).

Reading a q-value off the plot

The line from the **origin** through a point has slope $\boldsymbol{p_{(i)}/i}$. That **single slope** gives all **three** q-values: multiply it by each method's factor



$$\text{slope:} \quad \frac{p_{(152)}}{152} = \frac{0.01488}{152} = \mathbf{0.00009792}$$

BH	**BY**	**Storey**
factor $n = 1000$	factor $n\,c(n) = 7485$	factor $n\,\hat{\pi}_0 = 796$
$\boldsymbol{q = 0.098}$	$\boldsymbol{q = 0.733}$	$\boldsymbol{q = 0.078}$

At $q = 0.10$ the same test is rejected by BH and Storey, not by BY

Simulations and practicals

Shiny web applications

https://bioinformatics.cruk.cam.ac.uk/stats/shinystats/